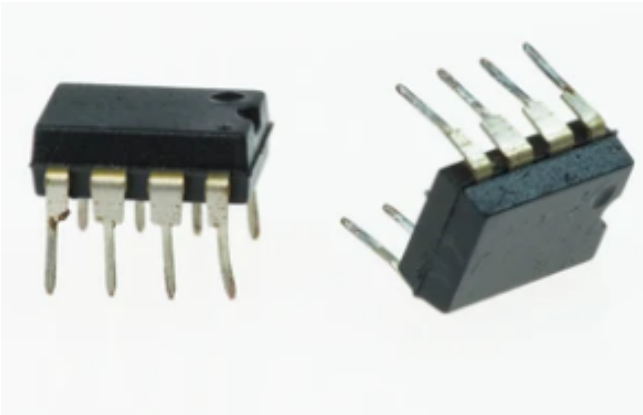
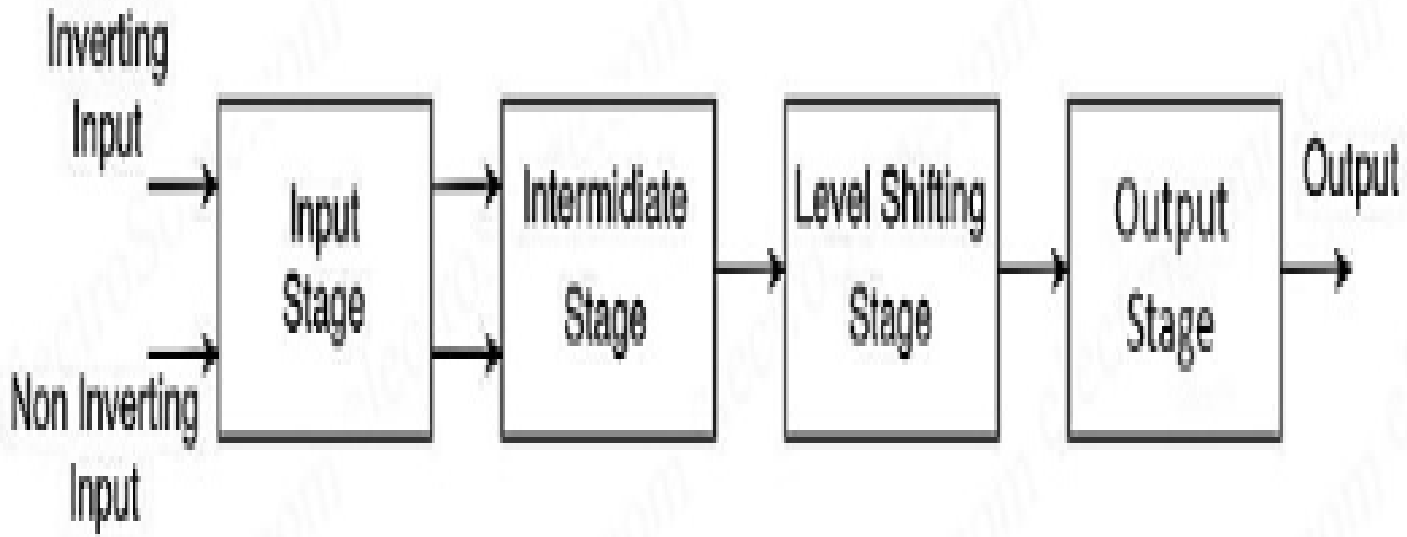
Op-Amp



Block diag of OP-AMP



OPAMP is basically a differential amp i.e., it will amplify the voltage which is differentially present between its i/p terminals

Op-amp pin diagram

There are 8 pins in a common Op-Amp, like the 741 which is used in many instructional courses.

Pin 1: Offset null
Pin 2: Inverting input terminal
Pin 3: Non-inverting input terminal
Pin 4: –VCC (negative voltage supply)
Pin 5: Offset null
Pin 6: Output voltage
Pin 7: +VCC (positive voltage supply)
Pin 8: No Connection

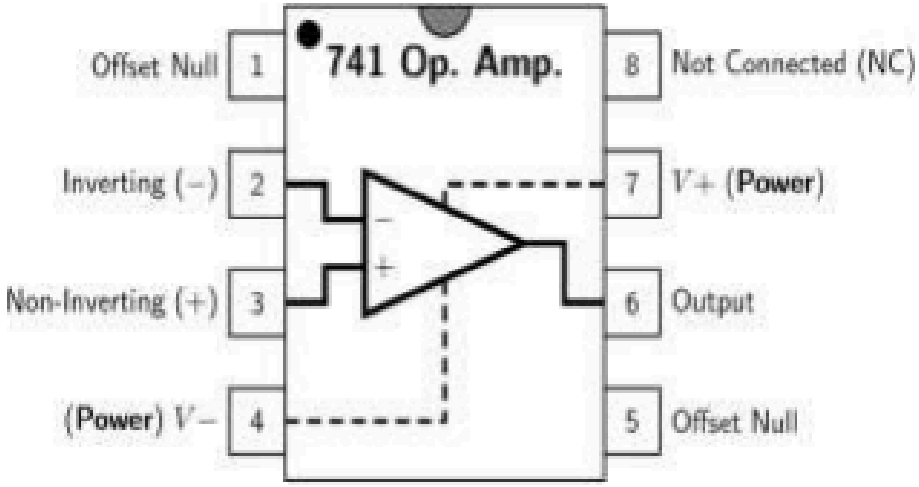


Figure : Pin connection, LM741.

- **Input bias current(I_b):** average of current that flows into the inverting and non-inverting input terminal of op-amp.
- **I/p and o/p impedance:** It is the resistance offered by the inputs and the output terminals to varying voltages. The quantity is expressed in Ohms.
- **Open Loop Gain:** It is the overall voltage gain or the amplification.
- **Input offset voltage :** It is a voltage that must be applied between the two terminal of an op-amp to null the o/p.
- **Input offset current (I_i):** The algebraic different between the current in to the inverting and Non-inverting terminal.

Input stage: The i/p stage is a dual i/p, balanced o/p differential amp. The 2 i/p are inverting and non- inverting i/p terminals. This stage provides most of the voltage gain of the OP-AMP and decides the value of i/p resistance Ri.

Intermediate stage: This is usually another differential amp. The i/p stage drives the stage. The stage is a dual –i/p unbalanced o/p differential amp.

Level- shifting stage: Due to direct coupling used between the 1st 2 stages, the i/p of level shifting stage is an amplified signal with some non – zero dc level. Level shifting stage is used to bring dc level to zero volts with respect to gnd.

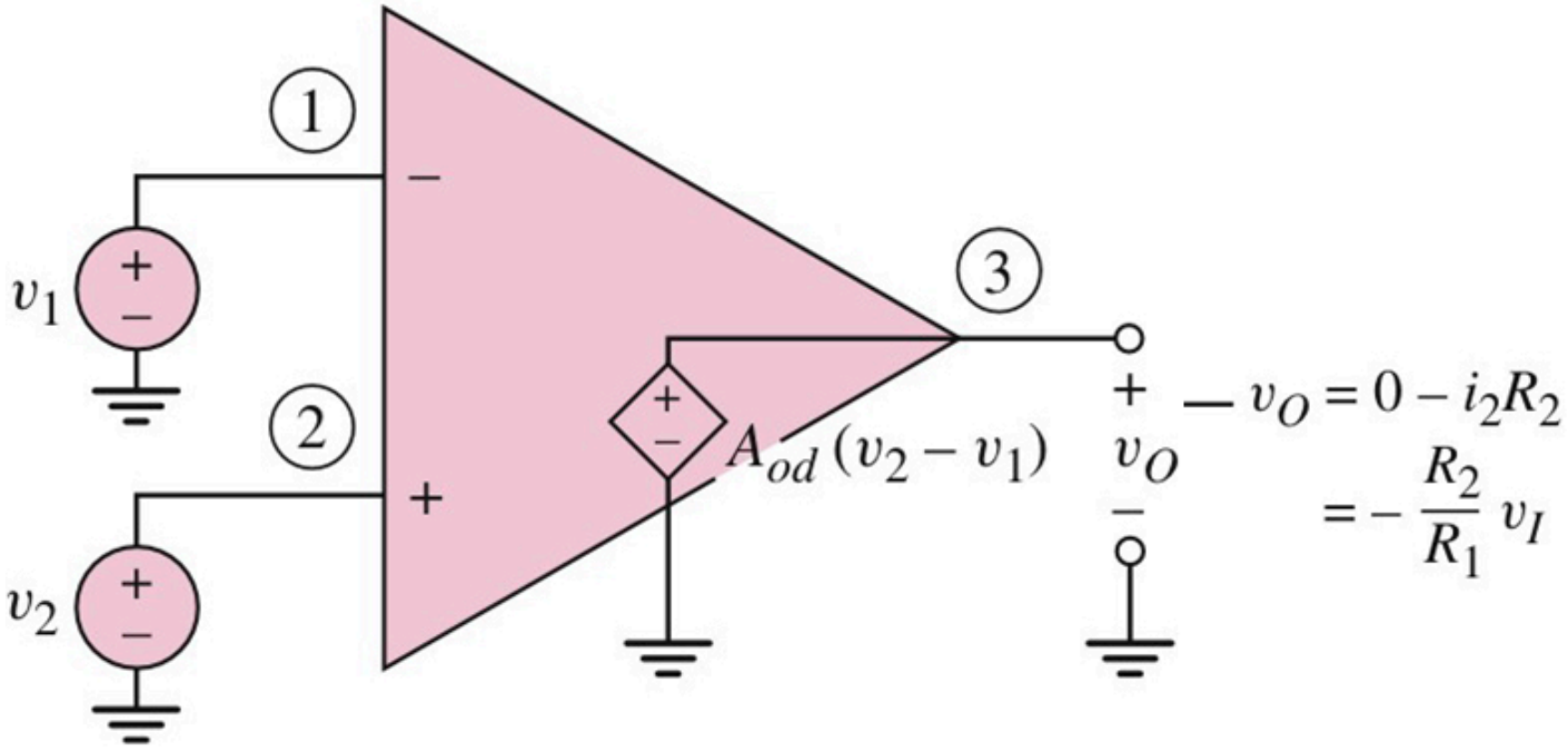
Output stage: (Emitter Follower)This stage is normally a complementary o/p stage. It increases the magnitude of voltage and raises the current supplying capability of OP-AMP. It also ensures that the o/p resistance of OPAMP is low.

Ideal Op-Amp

An ideal operational amplifier has the following characteristics:

1. An infinite voltage gain
2. An infinite bandwidth
3. An infinite input impedance
4. Zero output impedance
5. Perfect balance i.e., the output is zero when equal voltages are present at the two input terminals.
6. The characteristics do not change with temperature.

Ideal Op-Amp Equivalent Circuit

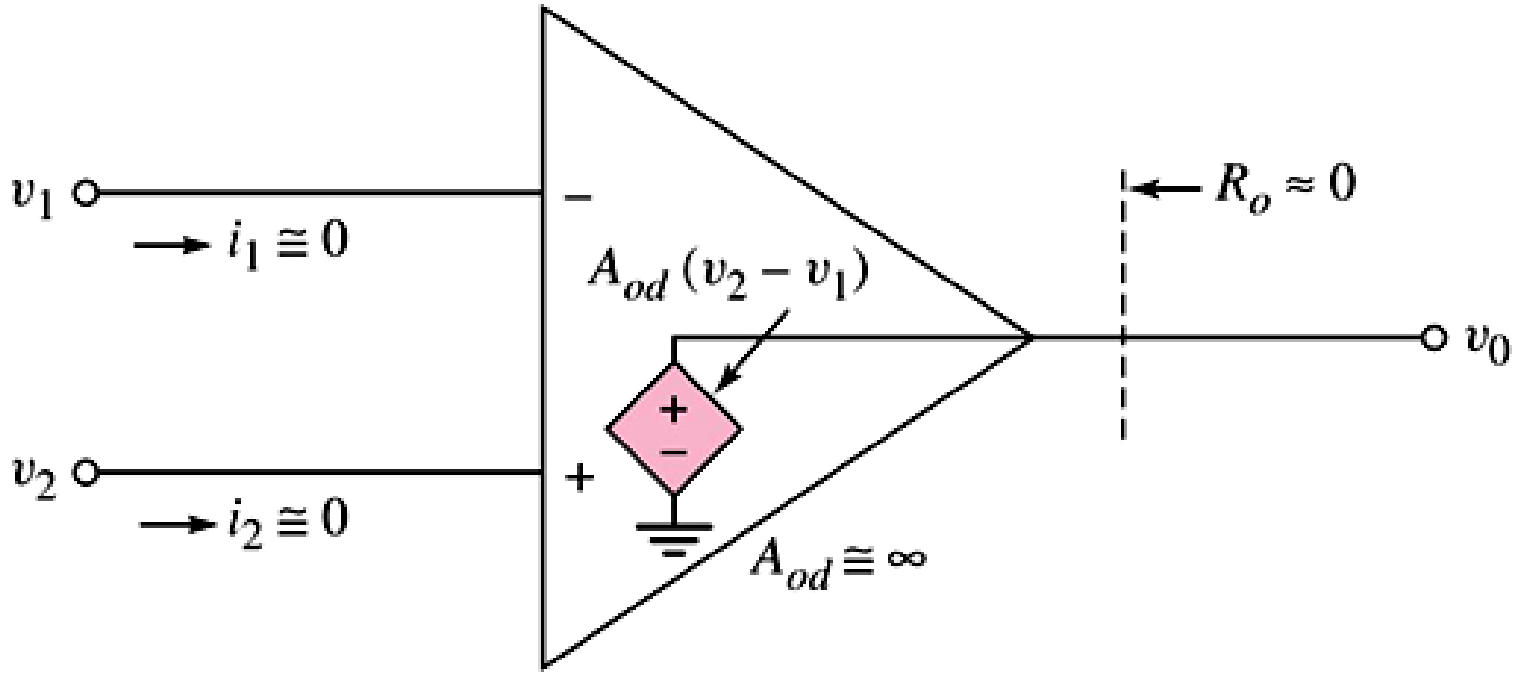


Characteristics of Ideal Op-Amp

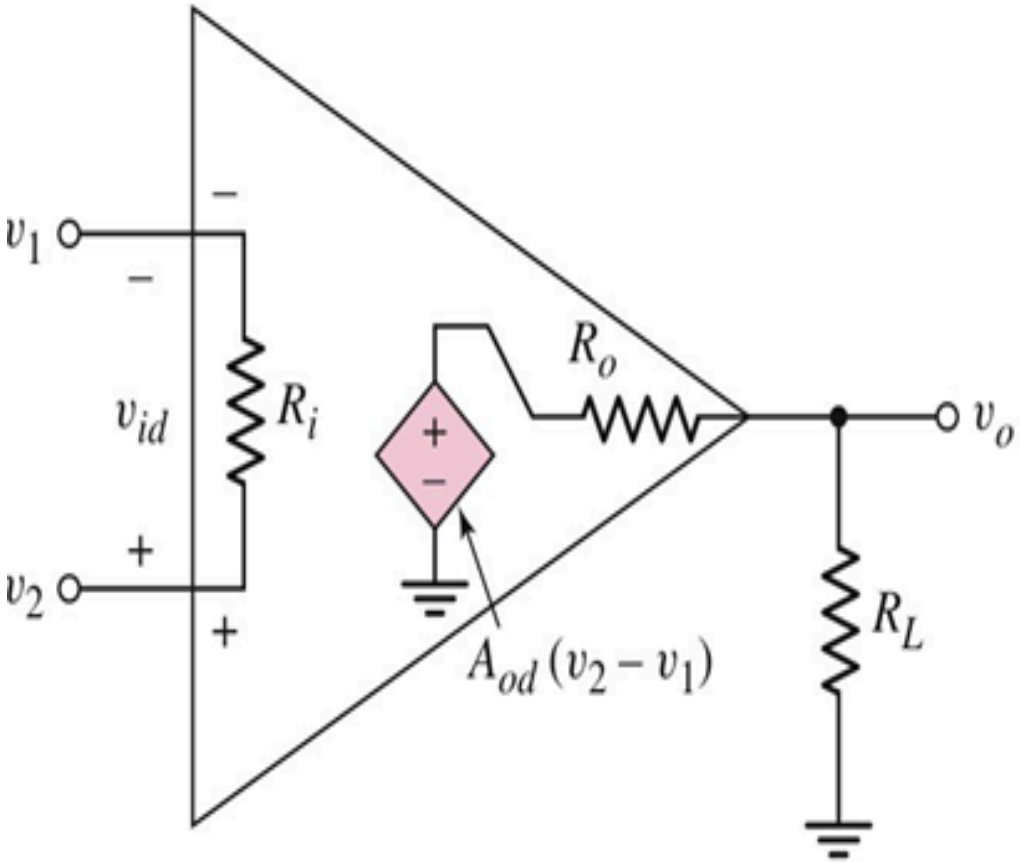
- Infinite input impedance
- Zero output impedance
- Infinite voltage gain
- Infinite bandwidth
- Infinite CMRR (Ad/Ac = Differential Gain/ Common mode gain)
- Infinite power supply rejection ratio

Ideal Op-Amp Characteristics

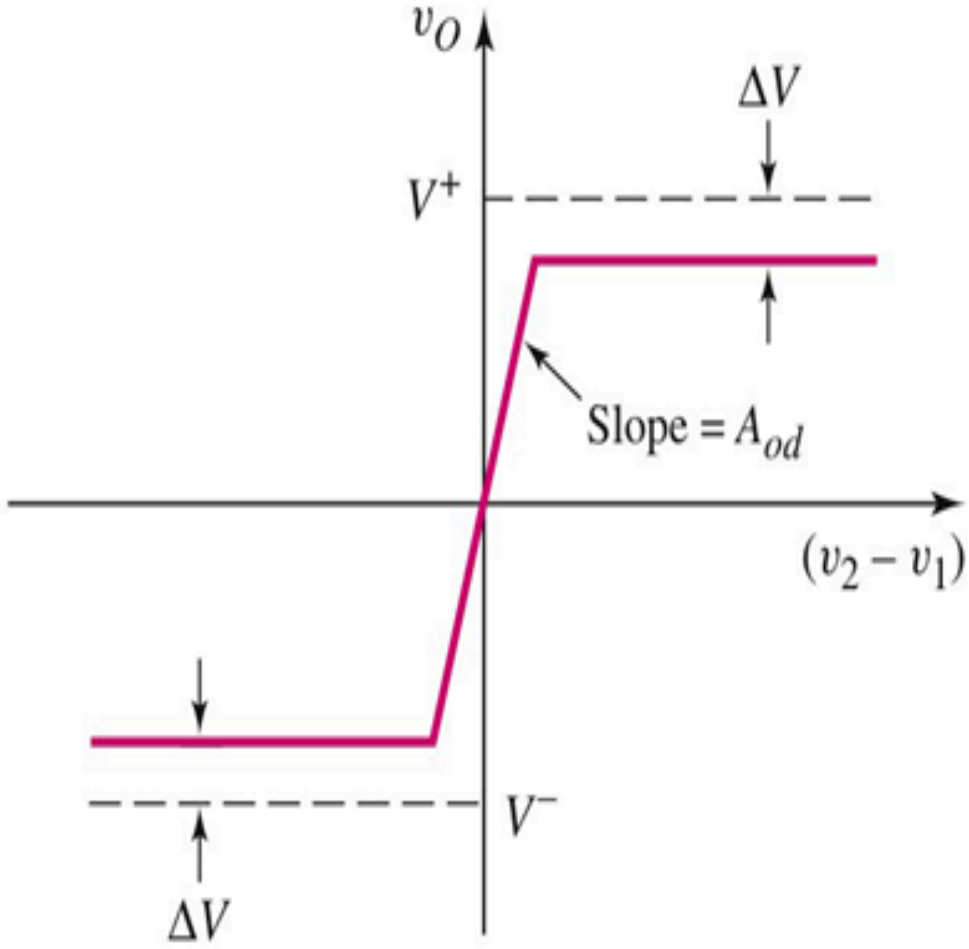
1. Internal differential gain A_{od} is infinite.
2. Differential input voltage (v_2-v_1) is zero.
3. Effective input resistance is infinite.
4. Output resistance is zero so output voltage is connected directly to dependent voltage source



Equivalent Circuit of Op-Amp

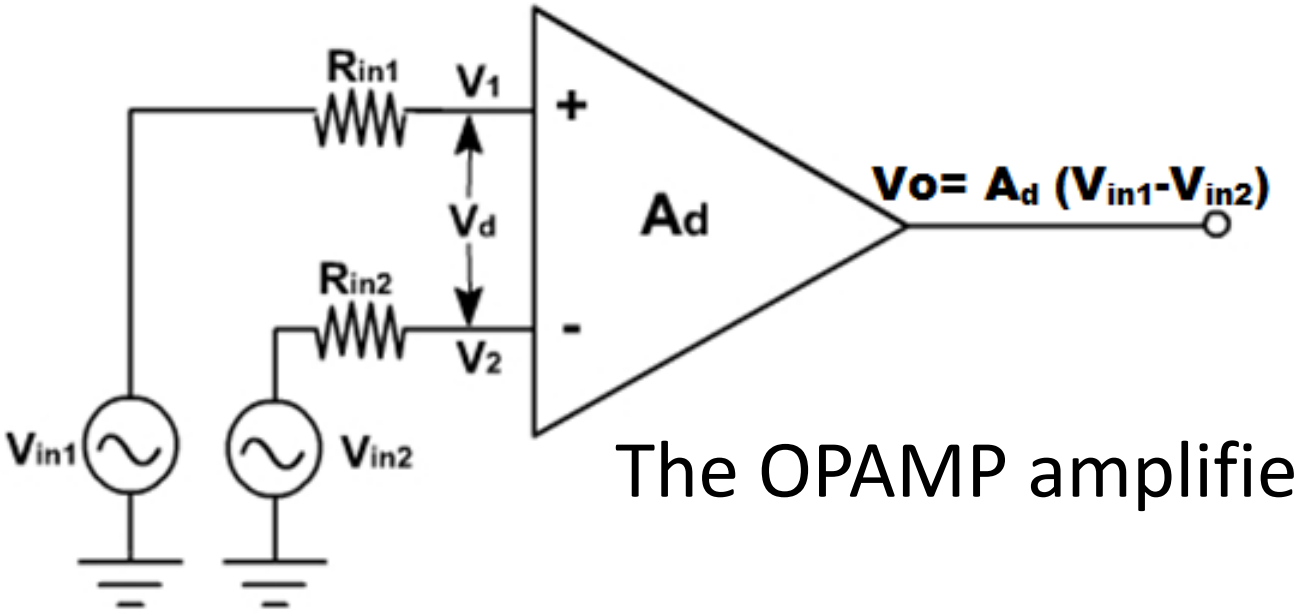


(a)



(b)

Open Loop : The Differential Amplifier



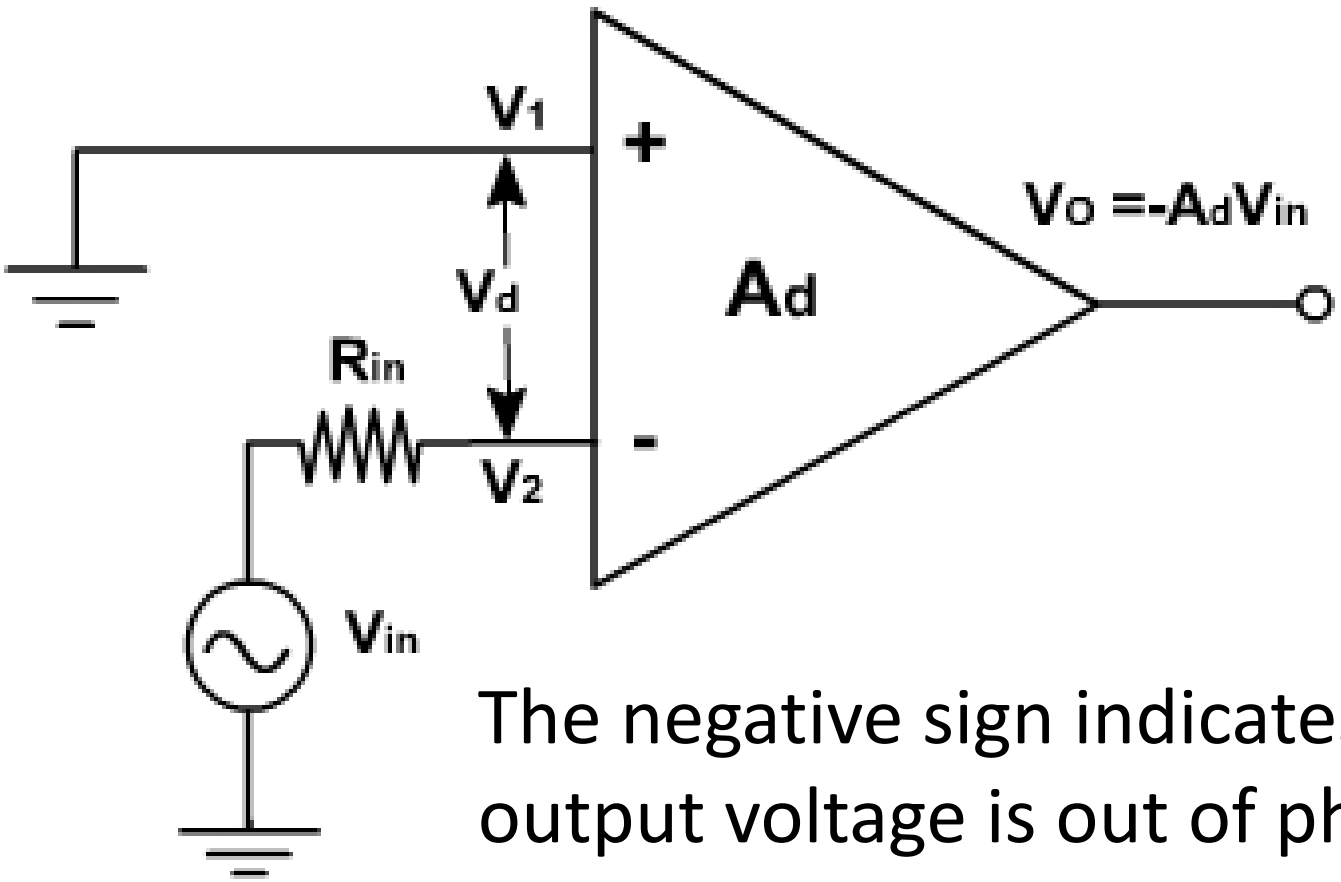
The OPAMP amplifies both ac and dc input signals. The source resistance R_{in1} and R_{in2} are normally negligible compared to the input resistance R_i.

Therefore voltage drop across these resistances can be assumed to be zero.

$v_1 = v_{in1}$ and $v_2 = v_{in2}$

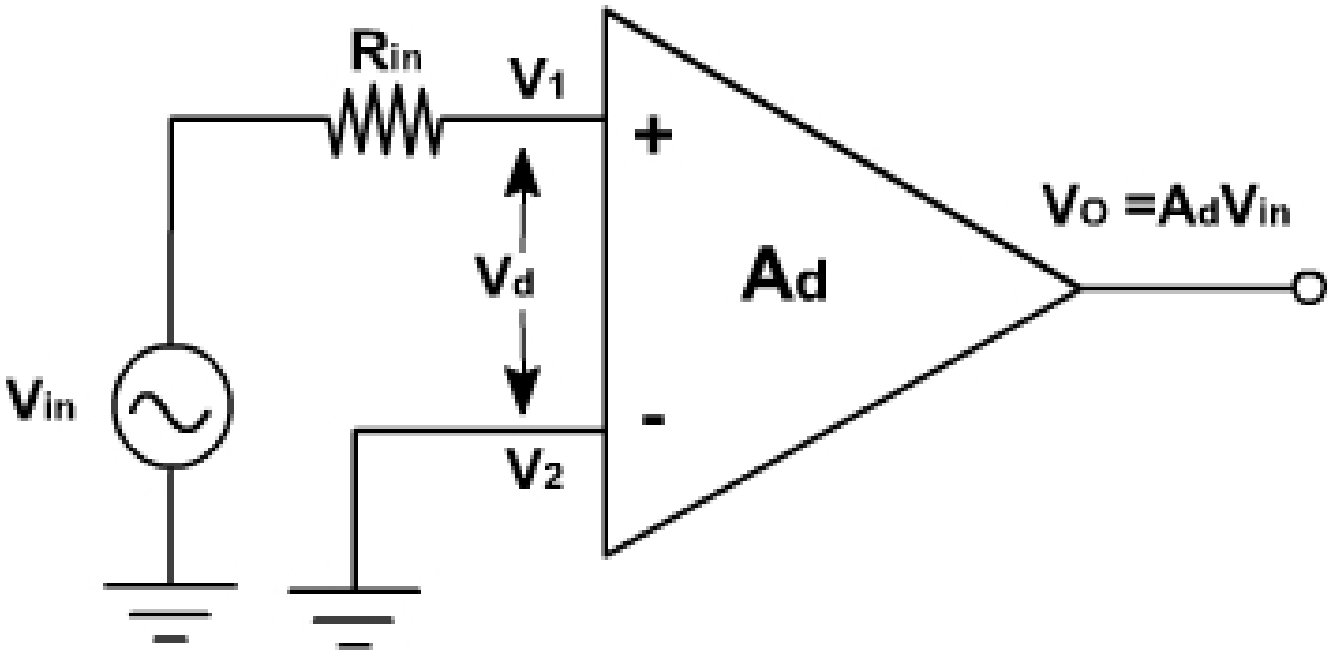
$v_o = A_d (v_{in1} - v_{in2})$ A_d is the open loop gain.

The Inverting Amplifier



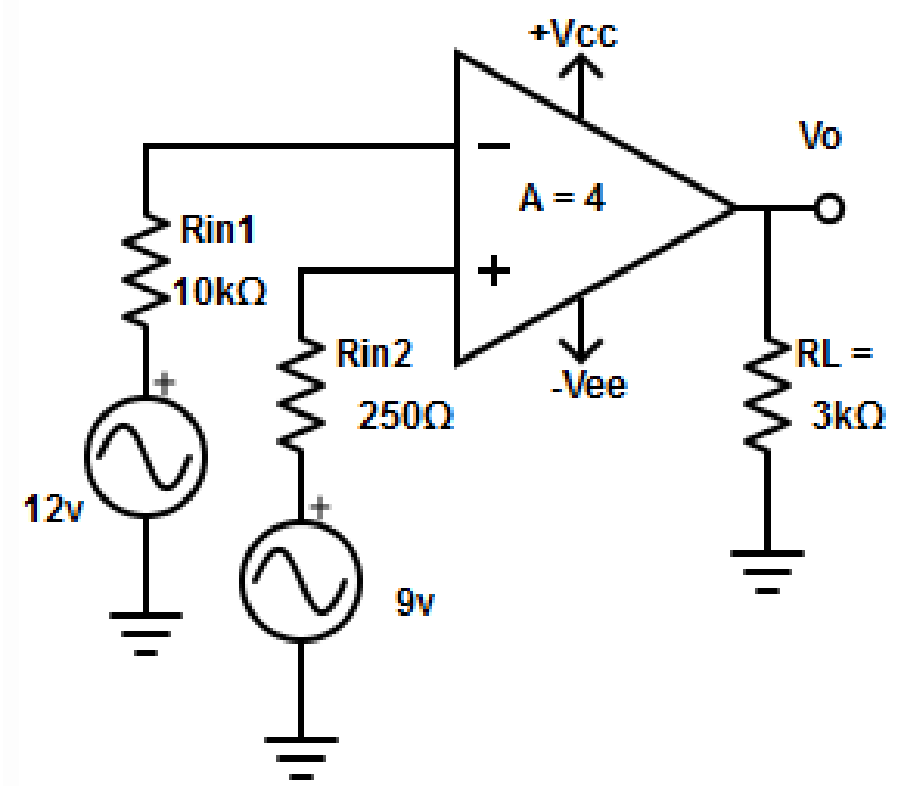
The negative sign indicates that the output voltage is out of phase with respect to input 180 ° or is of opposite polarity. Thus the input signal is amplified and inverted also.

The non-inverting amplifier



The input voltage is amplified by A_d and there is no phase reversal at the output.

Example



Calculate the output voltage for the given circuit.

Inverting amplifier

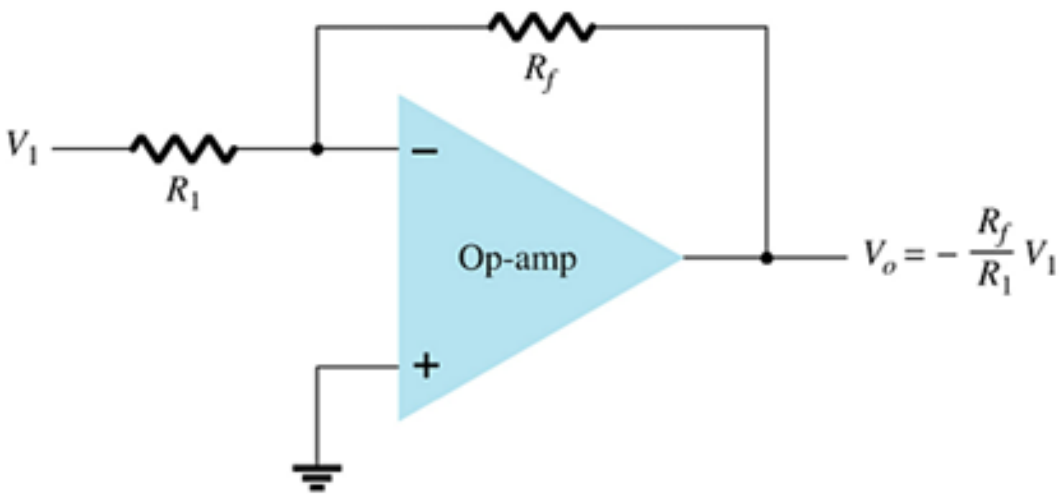
- Inverting amplifier is an amplifier whose amplified output is negatively proportional to the input.
- Its gain will be negative. For eg. if we provide sine wave input, its amplified output will be 180° out of phase with input.
- When the input is voltage is increasing, output voltage will decrease and vice versa.

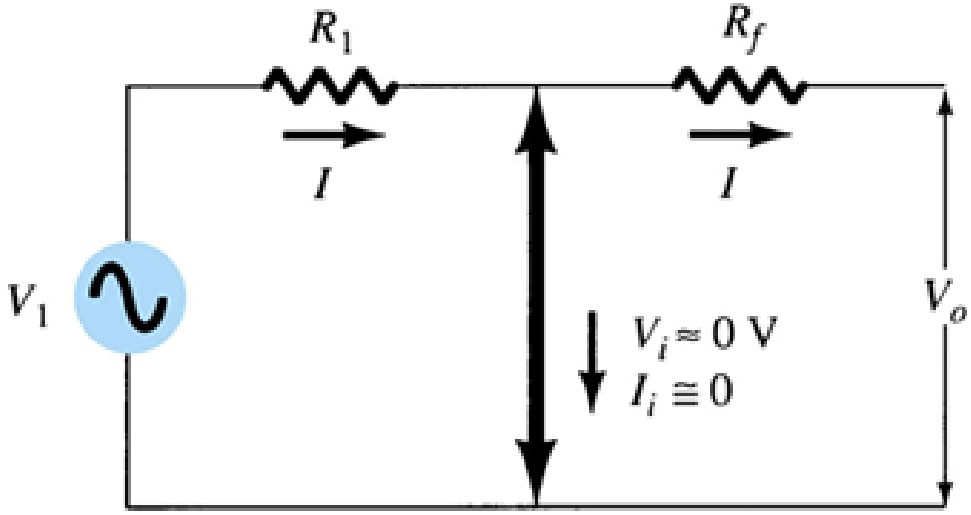
Virtual Ground

In op-amps **the term virtual ground means that the voltage at that particular node is almost equal to ground voltage** (0V).
It is **not** physically connected to ground.
This concept is very useful in analysis of op-amp circuits and it will make a lot of calculations very simple.

- Open loop gain of an operational amplifier is very high for practical applications.
- So we will apply negative feedback by connecting a suitable resistor between the output terminal and the inverting input to reduce the overall gain of the amplifier.
- In this we feedback a fraction of output to input terminal through a feedback network. Since we are giving this feedback to inverting input, it will reduce the overall gain of the circuit which is called as closed loop gain.
- We can easily control the closed loop gain by changing the feedback ratio.

Virtual Ground





- **An understanding of the concept of virtual ground provides a better understanding of how an op-amp operates.**
- **The *non-inverting* input pin is at ground. *The inverting* input pin is also at 0 V for an AC signal.**
- **The op-amp has such high input impedance that even with a high gain there is no current from inverting input pin, therefore there is no voltage from inverting pin to ground—all of the current is through R_f.**

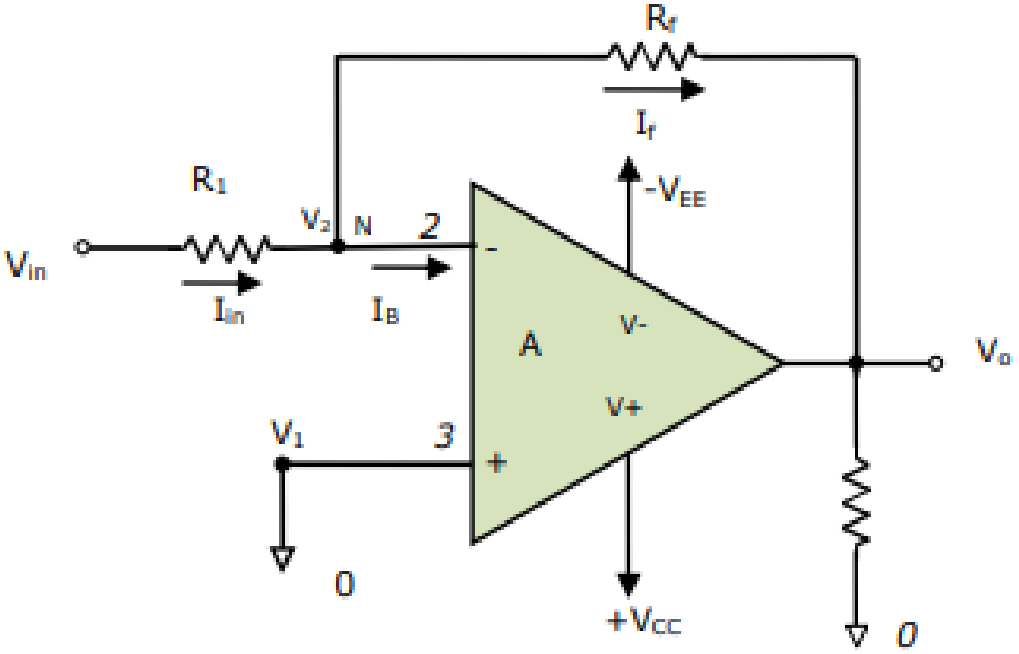
Two very important rules to remember about any operational amplifier

- No Current Flows into the Input Terminals
- The Differential Input Voltage is Zero as V1 = V2 = 0 (Virtual ground)
- Then by using these two rules we can derive the equation for calculating the closed-loop gain of an inverting amplifier.
- Concept is valid only when - **negative feedback** is applied to op-amp

Virtual Ground

Virtual Ground	Real Ground
Virtual Ground is a concept that made for easy explanation and calculation purposes.	Real Ground is a terminal which is physically connected to ground or earth which acts as the reference point for the entire circuit.
Voltage is **approximately** Zero	Voltage is Zero
Not able to sink infinite current	It is an infinite current sink
Not electrically connected to Ground	Electrically connected to Ground

Inverting Amplifier





Equivalent Circuit – Inverting Amplifier – Virtual Ground – Opamp

$$\frac{V_{in}-V_2}{R_1}=\frac{V_2-VO}{R_f} \qquad \text{----1}$$

Since $\frac{Vo}{A}$ = V_{id} = V_1-V_2

But V_1 is zero.

So V_2 = $-\frac{V_o}{A}$

$$\frac{V_{in}-\left(-\frac{Vo}{A}\right)}{R_1}=\frac{\left(-\frac{Vo}{A}\right)-VO}{R_f}$$

$$\frac{V_{in}A+V_o}{R_1}=\frac{-V_O-V_OA}{R_f}$$

$V_{in}A$ R_f = $-V_OR_1-V_OA$ R_1-V_o R_f = V_O (- R_f -AR_1-R_1)

...Inverting Amplifier

- $V_{in}A$ R_f = – $V_O R_1 - V_O A\ R_1$-Vo R_f

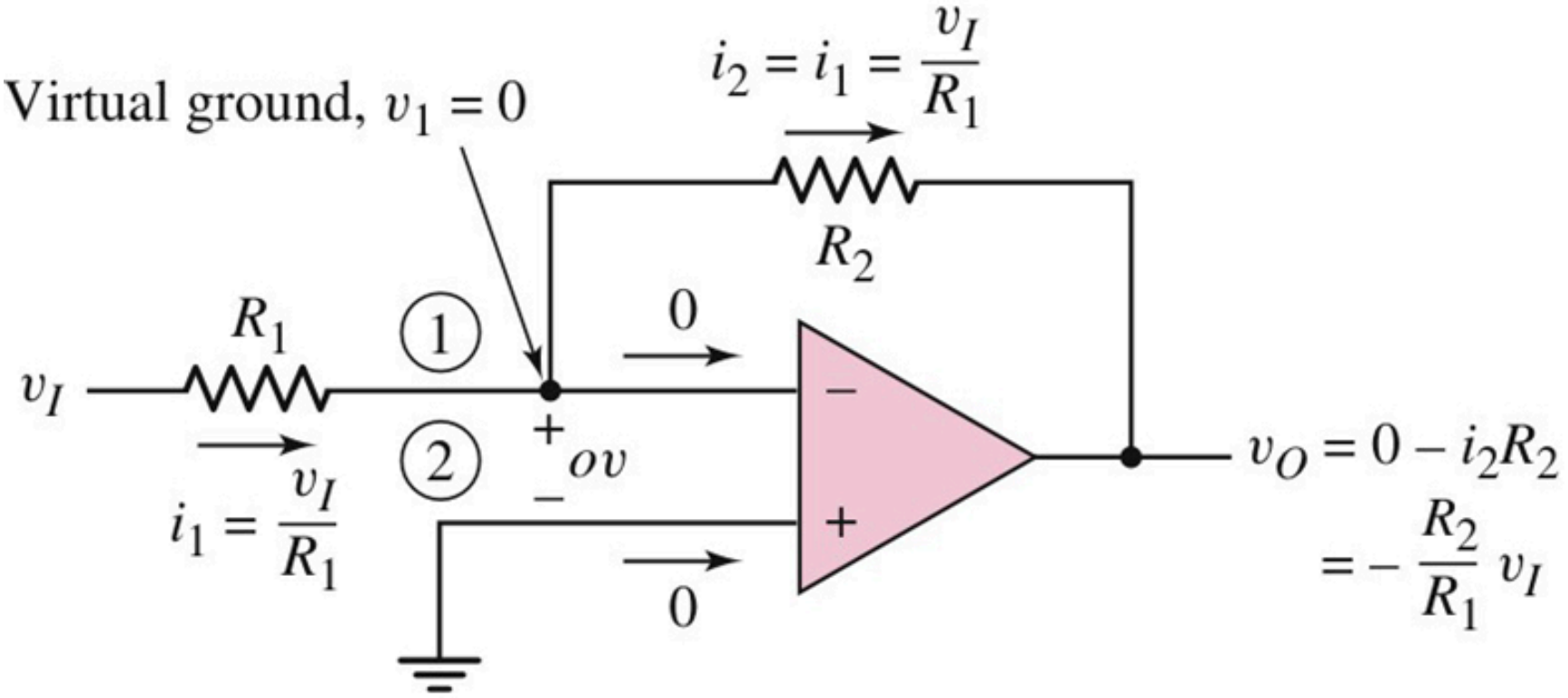
  = V_O (- R_f -AR_1-R_1)

Vo/Vin = A R_f /(- R_f -AR_1-R_1)

= - A R_f /(R_f +AR_1+R_1) ($AR_1 \gg R_f + R_1$)

A_f = - R_f / R_1

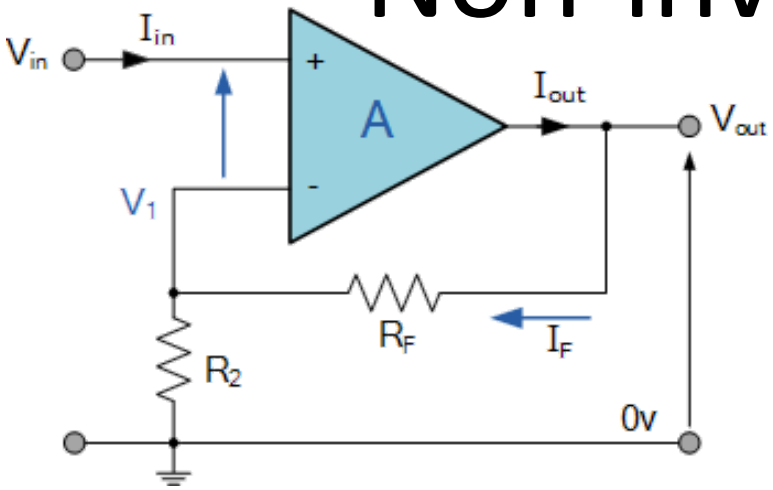
Inverting Amplifier



$A = -R_2/R_1$

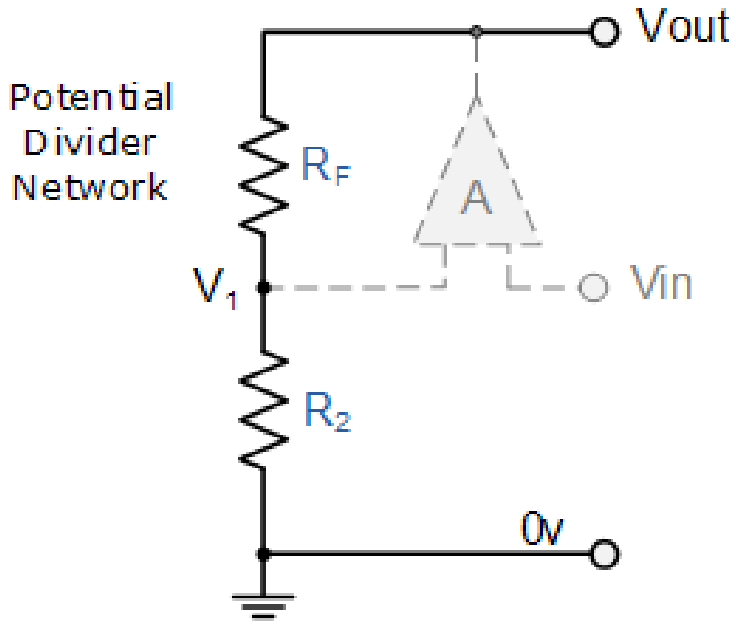
$= -R_F/R_1$

Non-inverting amplifier



$$V_1 = \frac{R_2}{R_2 + R_F} \times V_{OUT}$$

Ideal Summing Point: $V_1 = V_{IN}$



Voltage Gain, $A_{(V)}$ is equal to: $\frac{V_{OUT}}{V_{IN}}$

Then, $A_{(V)} = \frac{V_{OUT}}{V_{IN}} = \frac{R_2 + R_F}{R_2}$

Transpose to give: $A_{(V)} = \frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_F}{R_2}$

Voltage Follower (Unity Gain Buffer)

The ideal op-amp rules are:

- The differential input voltage is zero. $Vi = 0 \rightarrow Vn = Vp$
- No current flowing into the input terminals. $In = Ip = 0$ This is equivalent to infinite input resistance for the op-amp Ri= ∞
- Infinite open loop gain. $A \rightarrow \infty$
- Output resistance is zero $Ro = 0$

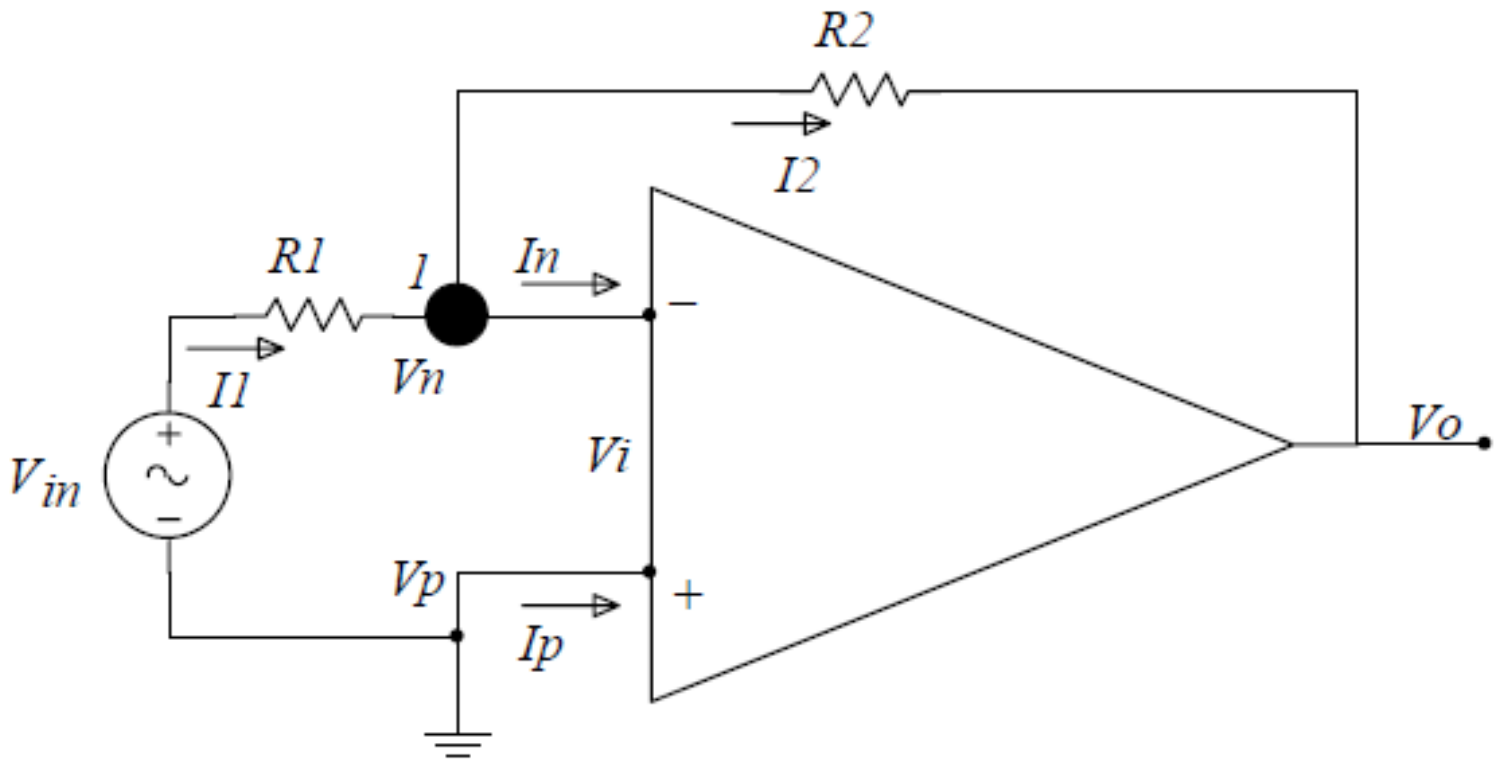


Figure 12. Ideal op-amp inverting amplifier circuit.

- Since *In=Ip=0*, KCL at node 1 tells us that current I_1 must be equal to current I_2.

$$I_1 = \frac{V_{in} - V_1}{R1} = \frac{V_{in1}}{R1} = I_2$$

$$I2 = \frac{Vn - Vo}{R2} \Rightarrow Vo = -I2\,R2 = -Vin\frac{R2}{R1}$$

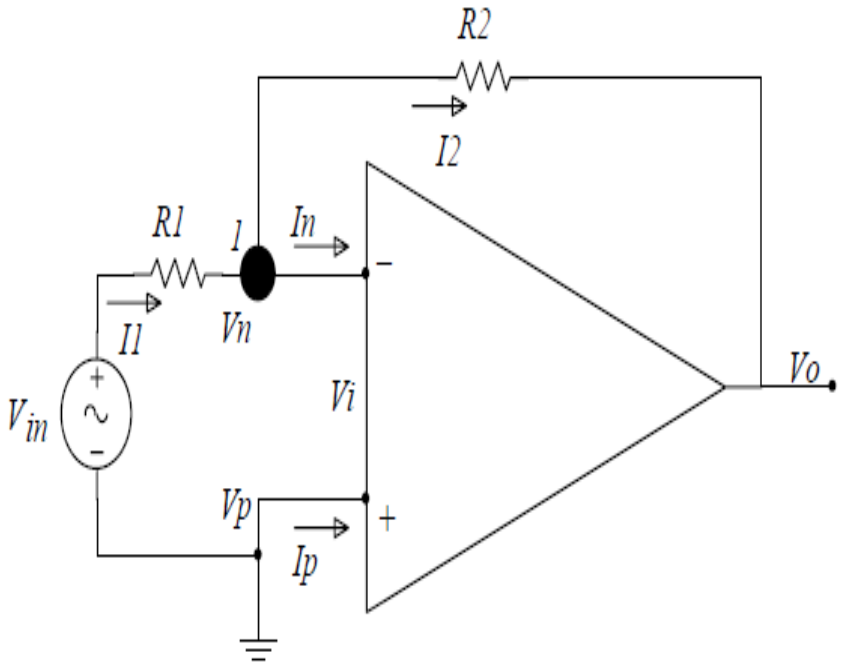
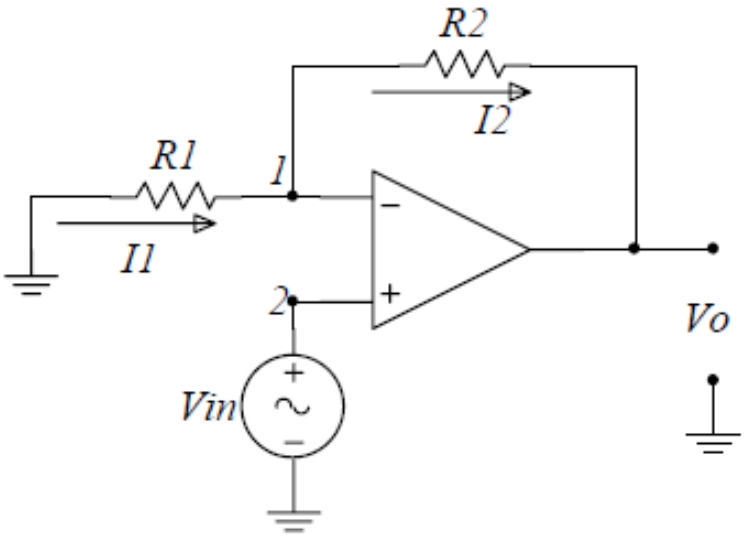


Figure 12. Ideal op-amp inverting amplifier circuit.

Gain of ideal inverting amplifier

$$Gain = \frac{V_o}{V_{in}} = \frac{-R_2}{R_1}$$

Noninverting Amplifier



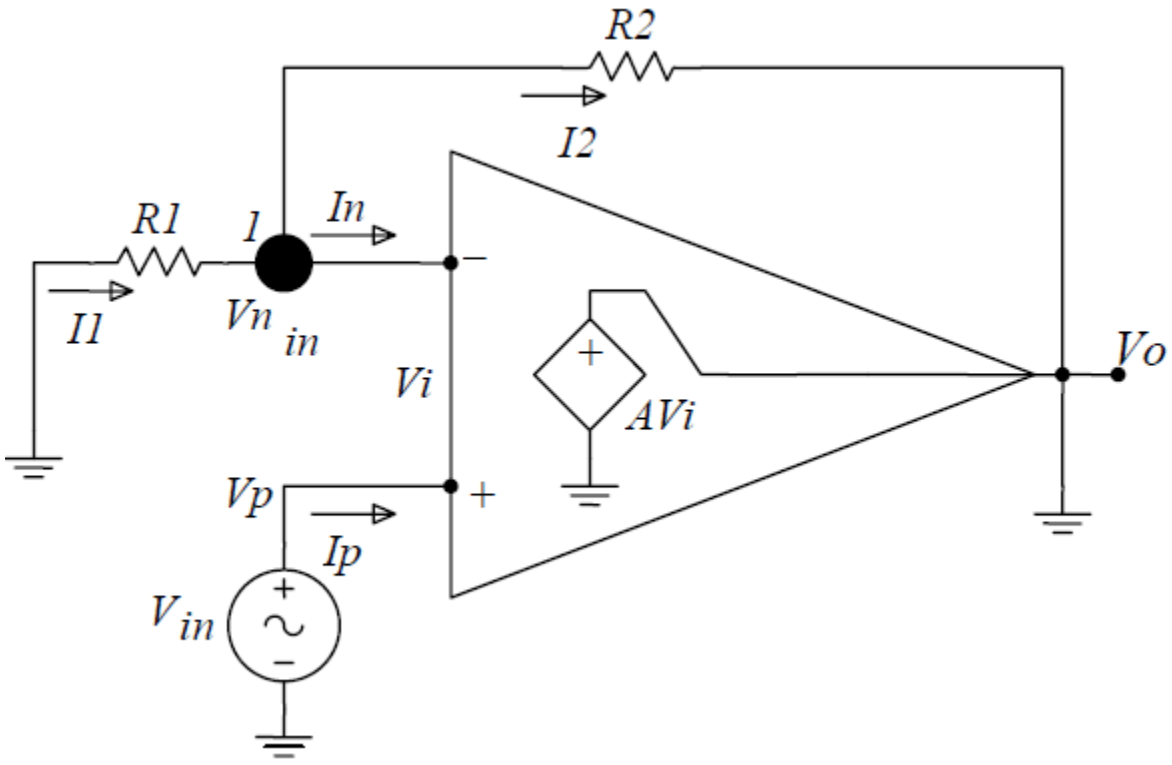


Figure 13. Non-inverting amplifier

Since $In = Ip=0$, we have $I1=I2$ and therefore:

$$\frac{-Vn}{R1}=\frac{Vn-Vo}{R2}\Rightarrow\frac{Vo}{R2}=Vn\left(\frac{1}{R1}+\frac{1}{R2}\right)$$

Since the voltage $Vi = Vp\text{-}Vn = Vin\text{-}Vn$, the output voltage is given by:

Since $In = Ip=0$, we have $I1=I2$ and therefore:

$$\frac{-Vn}{R1}=\frac{Vn-Vo}{R2}\Rightarrow\frac{Vo}{R2}=Vn\left(\frac{1}{R1}+\frac{1}{R2}\right)$$

Since the voltage $Vi = Vp\text{-}Vn = Vin\text{-}Vn$, the output voltage is given by:

- $Vo = A\,(Vin - Vn)$

$$G\equiv\frac{Vo}{Vin}=\frac{1+R2/R1}{1+(1+R2/R1)/A}$$

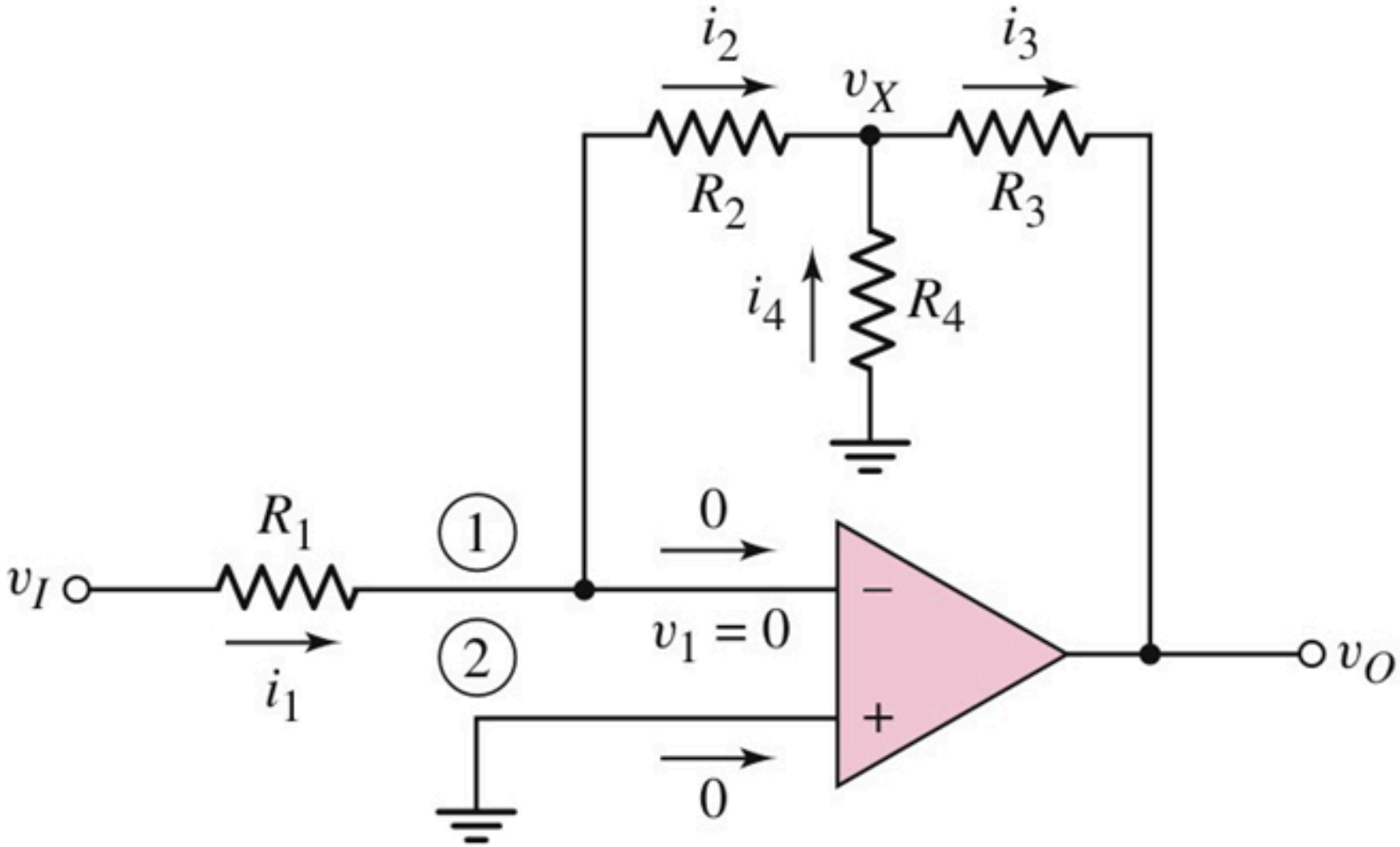
$A \to \infty$, the closed loop gain

$$\underset{A\to\infty}{G}=1+\frac{R2}{R1}$$

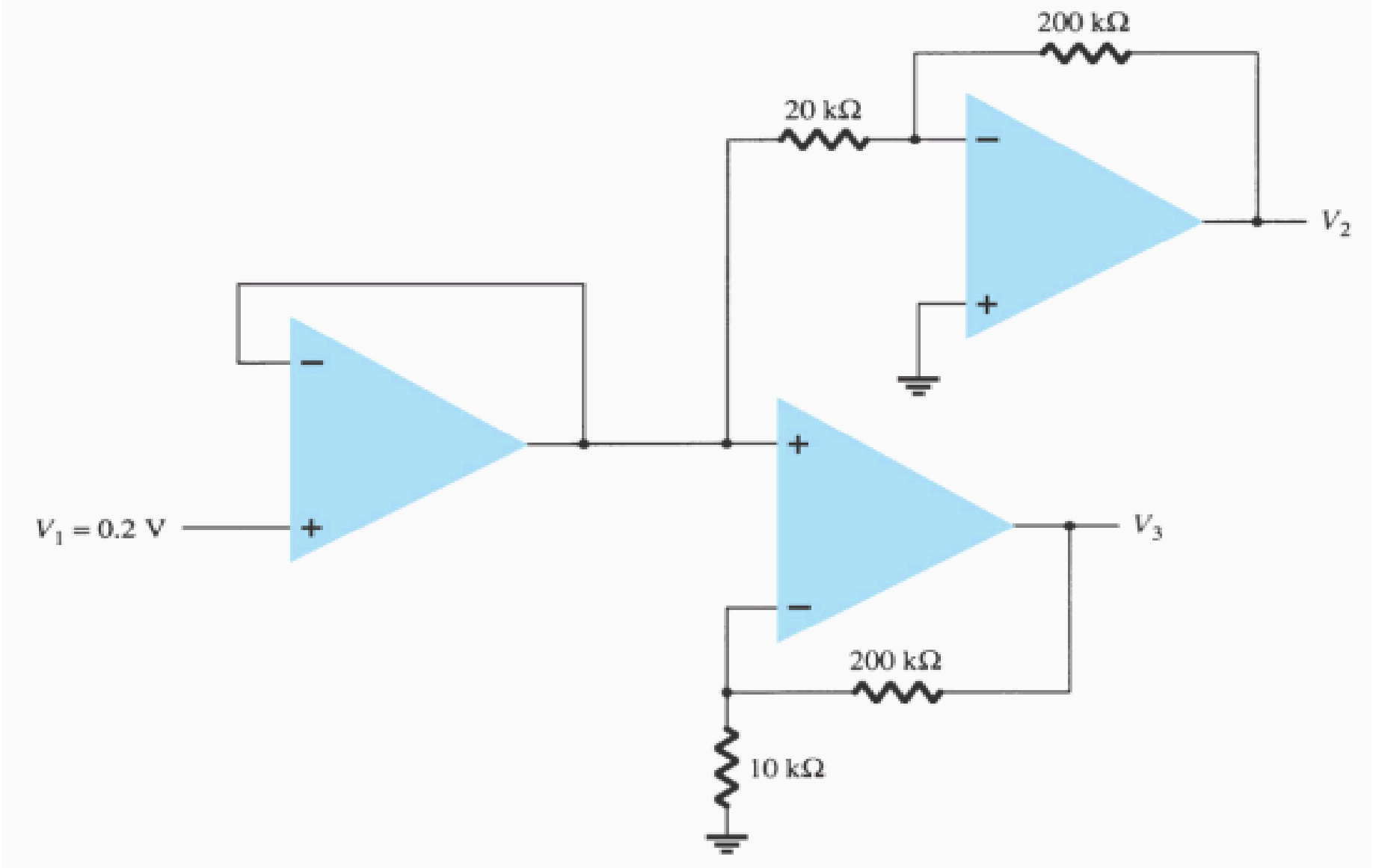
Problem-Solving Technique: Ideal Op-Amp Circuits

1. If noninverting terminal is grounded, then inverting terminal is virtual ground.
   a. Sum currents at node assuming no current enters Op Amp.
2. If noninverting terminal is not grounded, then inverting terminal voltage is equal to that of the noninverting terminal.
   a. Sum currents at node assuming no current enters Op Amp.
3. Output voltage is determined from either Step 1 or 2.

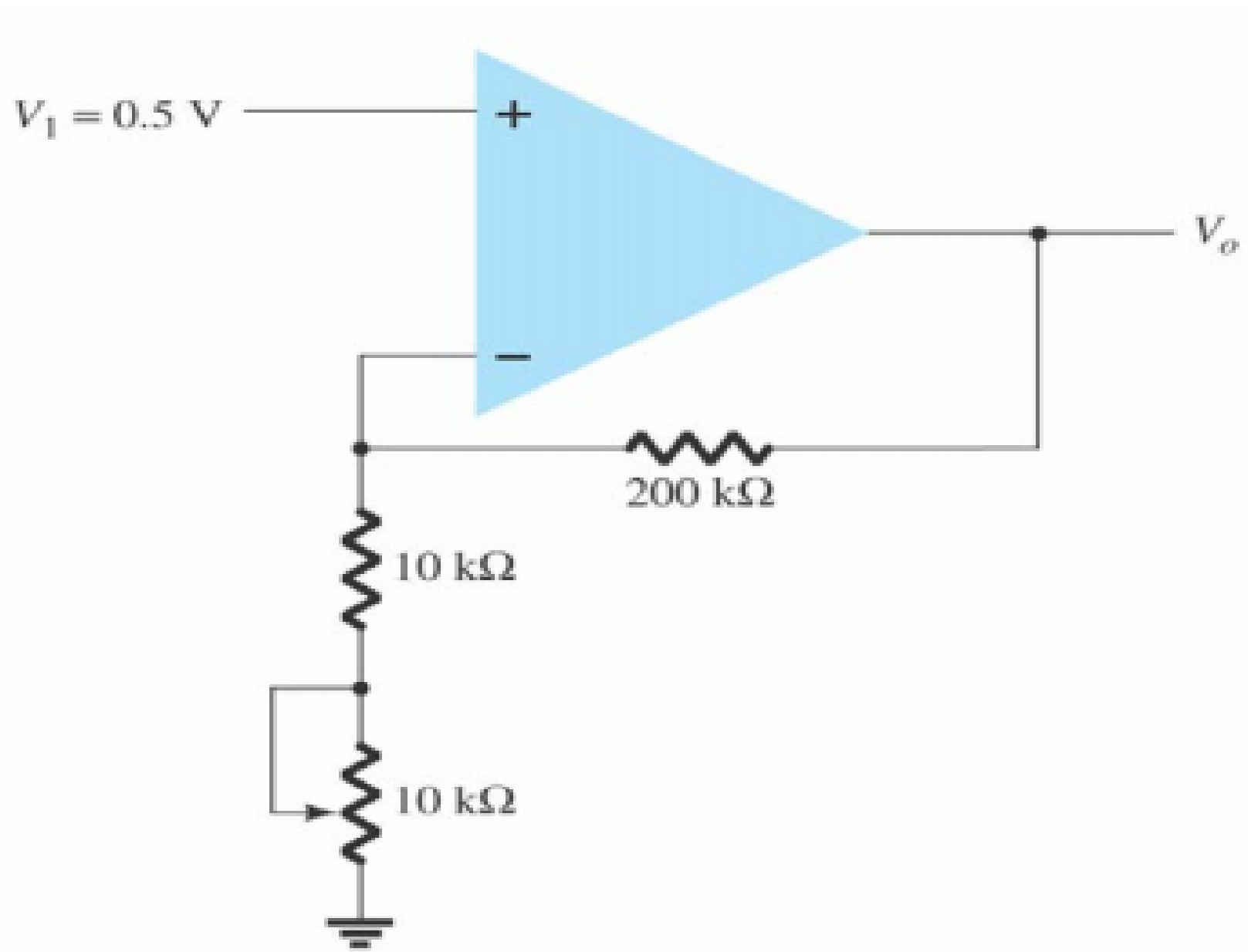
Calculate voltage gain Av



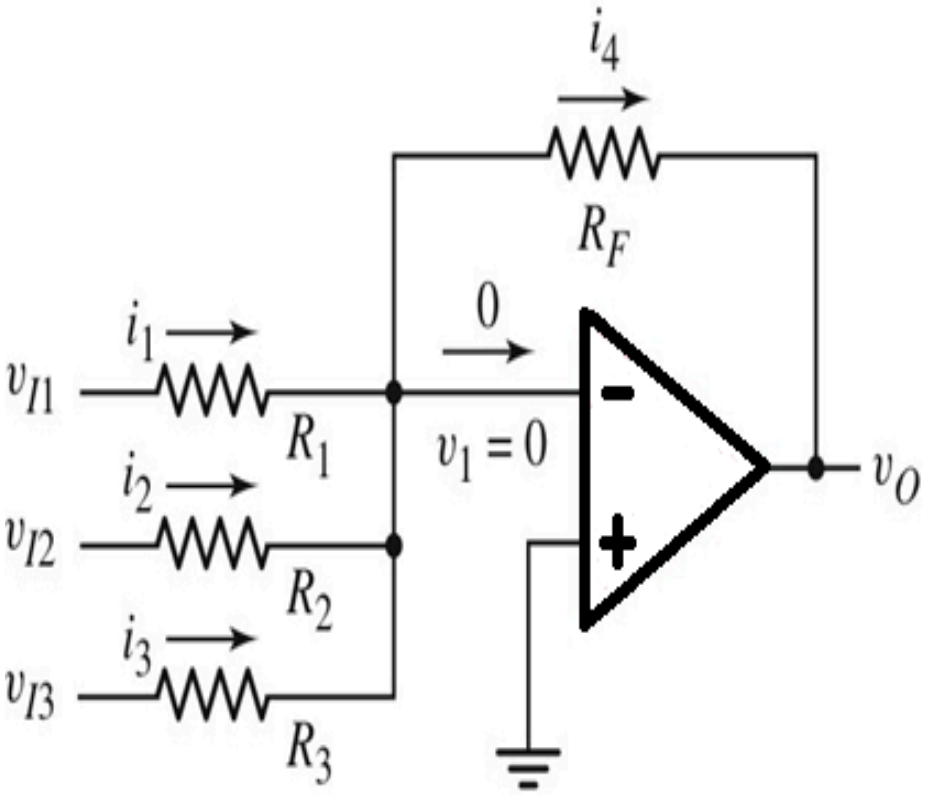
Calculate the output voltages V_2 and V_3



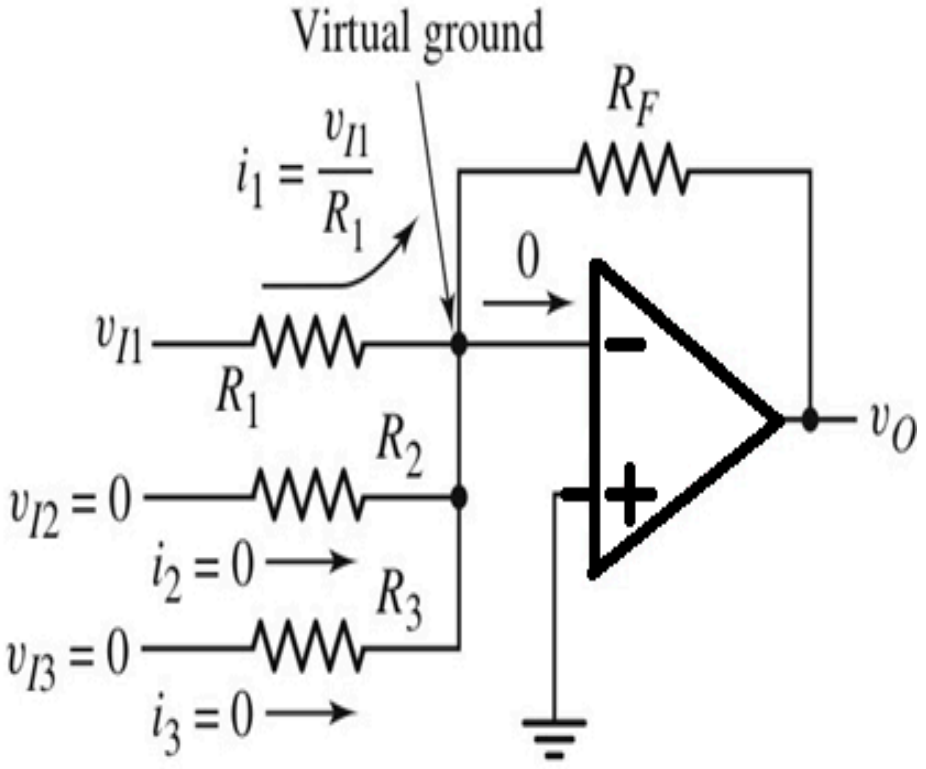
Calculate range of output voltage.



Adder-Summing Amplifier

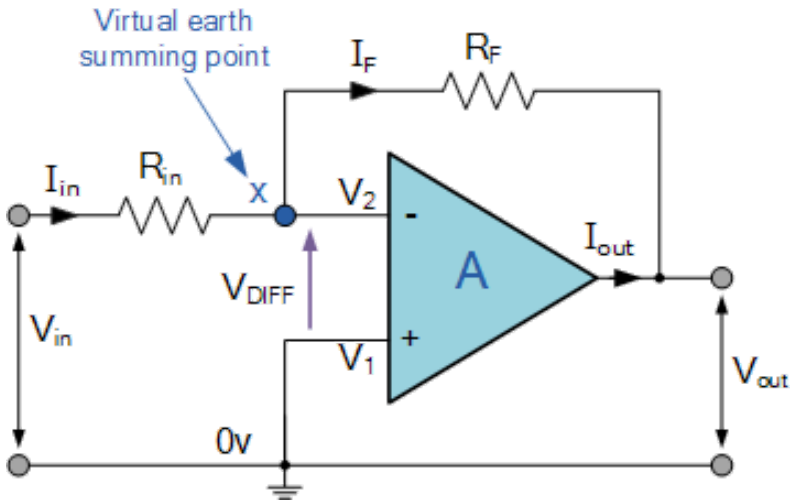


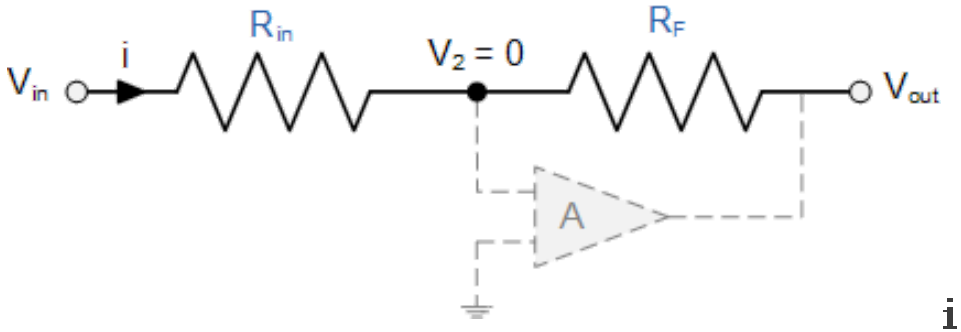
(a)



(b)

- In this **Inverting Amplifier** circuit the operational amplifier is connected with feedback to produce a closed loop operation. When dealing with operational amplifiers rules to remember about inverting amplifiers, these are:
- 1.“No current flows into the input terminal”
- 2. “V1 always equals V2”.





$$i = \frac{Vin - Vout}{Rin + Rf}$$

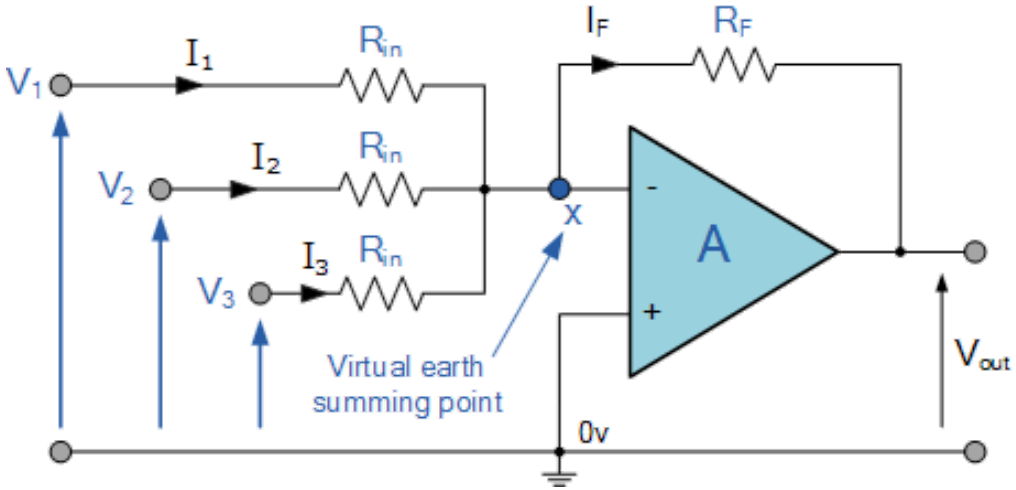
$$\text{therefore, } i = \frac{Vin - V2}{Rin} = \frac{V2 - Vout}{Rf}$$

$$i = \frac{Vin}{Rin} - \frac{V2}{Rin} = \frac{V2}{Rf} - \frac{Vout}{Rf}$$

$$\text{so, } \frac{Vin}{Rin} = V2\left[\frac{1}{Rin} + \frac{1}{Rf}\right] - \frac{Vout}{Rf}$$

$$\text{and as, } i = \frac{Vin - 0}{Rin} = \frac{0 - Vout}{Rf} \qquad \frac{Rf}{Rin} = \frac{0 - Vout}{Vin - 0}$$

$$\text{the Closed Loop Gain (Av) is given as, } \frac{Vout}{Vin} = -\frac{Rf}{Rin}$$

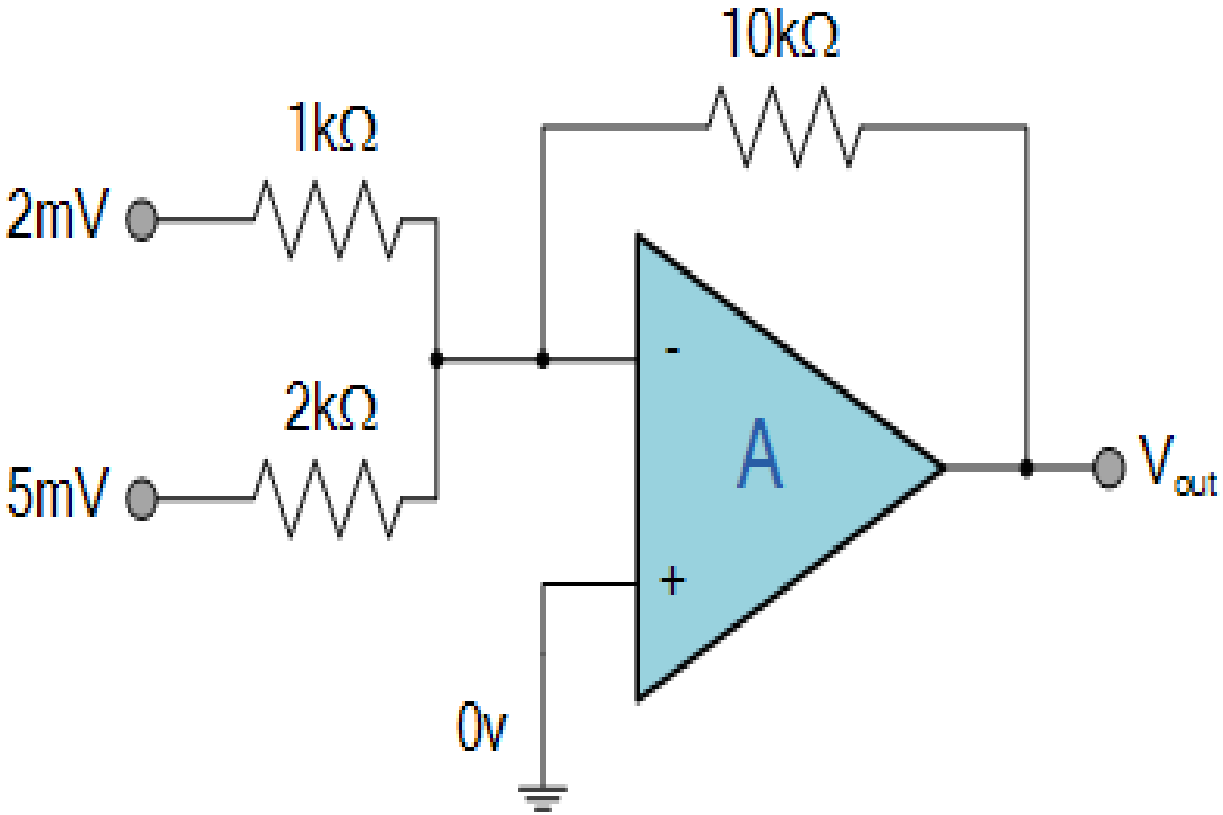


$$I_F = I_1 + I_2 + I_3 = -\left[\frac{V1}{Rin} + \frac{V2}{Rin} + \frac{V3}{Rin}\right]$$

$$\text{Inverting Equation: } Vout = -\frac{Rf}{Rin} \times Vin$$

$$\text{then, } -Vout = \left[\frac{R_F}{Rin}V1 + \frac{R_F}{Rin}V2 + \frac{R_F}{Rin}V3\right]$$

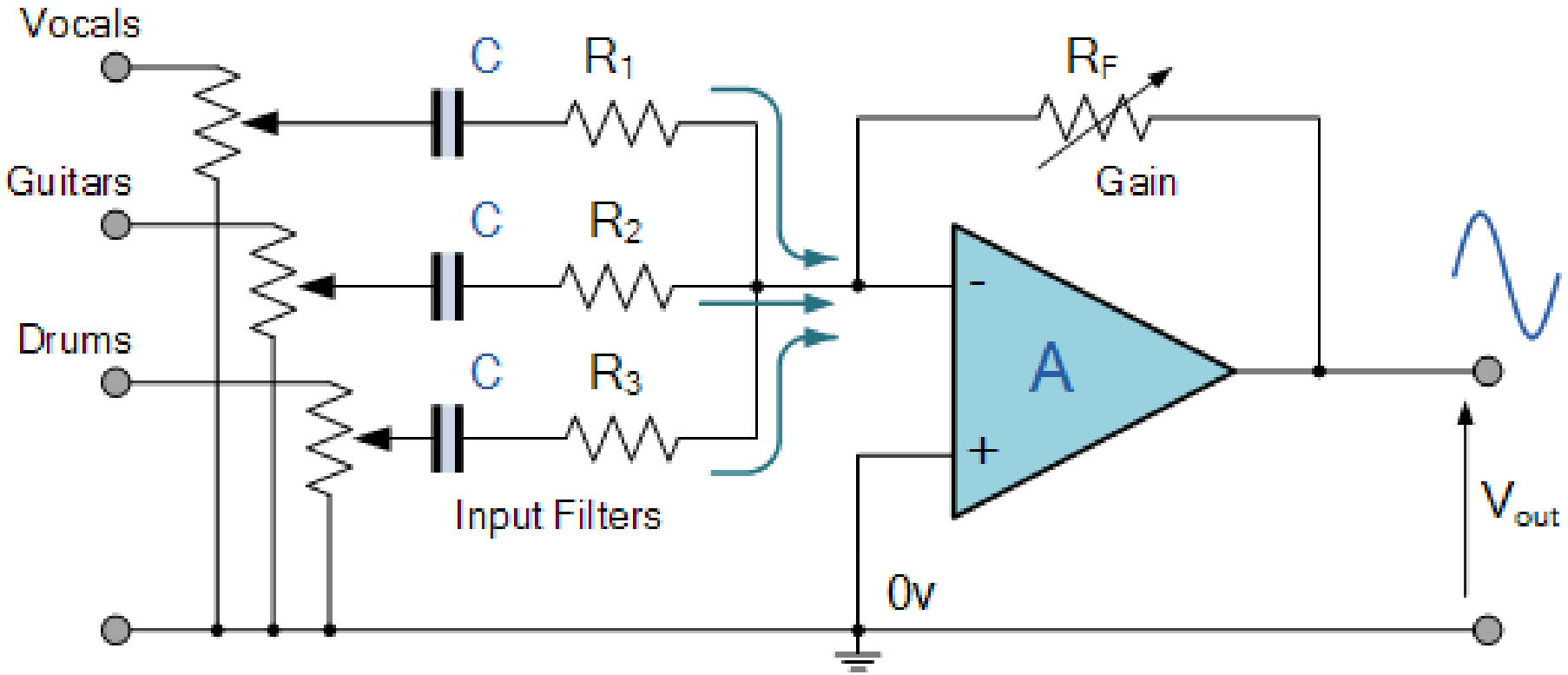
Find the output voltage of the following *Summing Amplifier* circuit

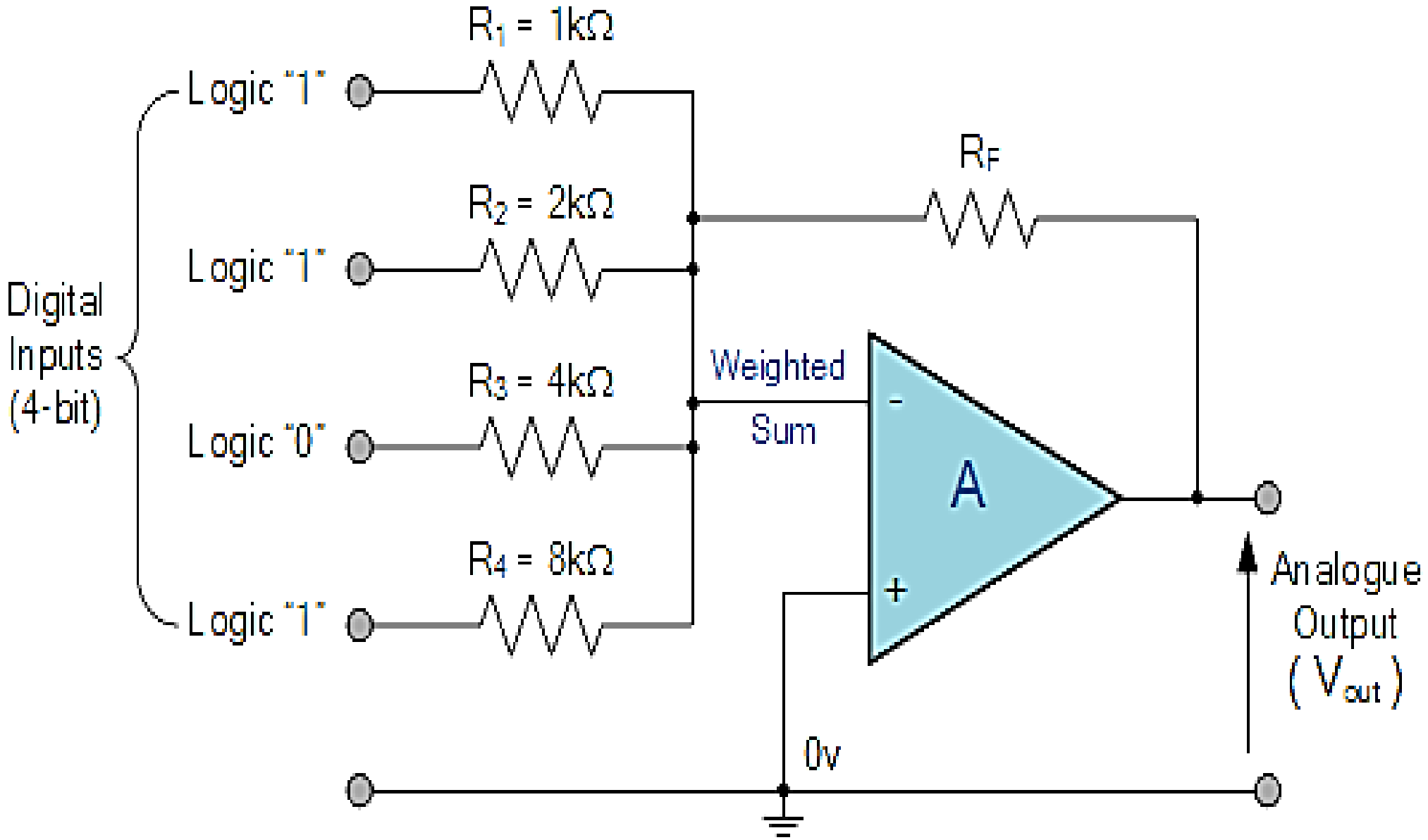


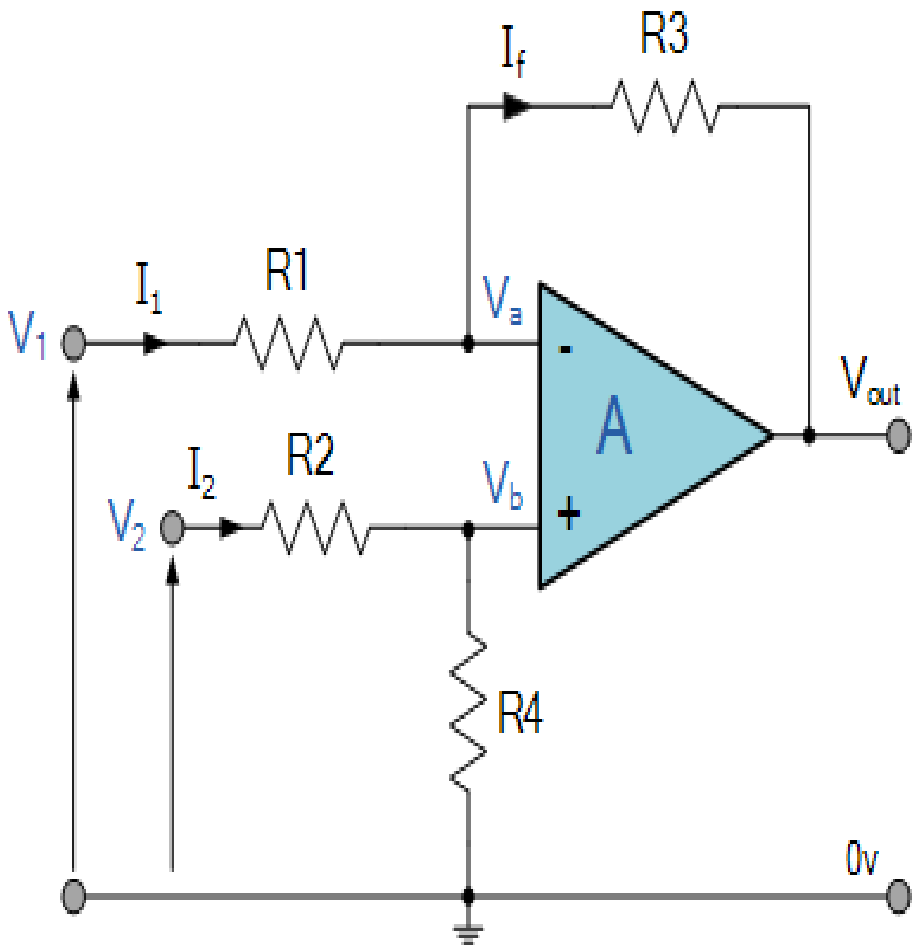
Vout = -45 mV

Scaling Summing Amplifier

$$-Vout = V1\left(\frac{Rf}{R1}\right) + V2\left(\frac{Rf}{R2}\right) + V3\left(\frac{Rf}{R3}\right) \ldots.etc$$




R1 = 1kΩ
Logic "1"
R2 = 2kΩ
Logic "1"
Digital
Inputs
(4-bit)
R3 = 4kΩ
Logic "0"
R4 = 8kΩ
Logic "1"
RF
Weighted
Sum
A
-
+
Analogue
Output
(Vout)
0v



$$I_1 = \frac{V_1 - V_a}{R_1}, \quad I_2 = \frac{V_2 - V_b}{R_2}, \quad I_f = \frac{V_a - (V_{out})}{R_3}$$

Summing point $V_a = V_b$

and $V_b = V_2\left(\frac{R_4}{R_2 + R_4}\right)$

If $V_2 = 0$, then: $V_{out(a)} = -V_1\left(\frac{R_3}{R_1}\right)$

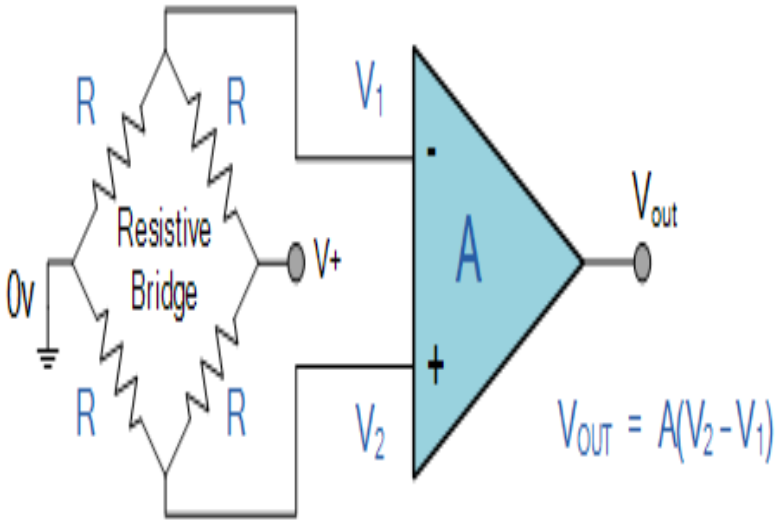
If $V_1 = 0$, then: $V_{out(b)} = V_2\left(\frac{R_4}{R_2 + R_4}\right)\left(\frac{R_1 + R_3}{R_1}\right)$

$$V_{out} = -V_{out(a)} + V_{out(b)}$$

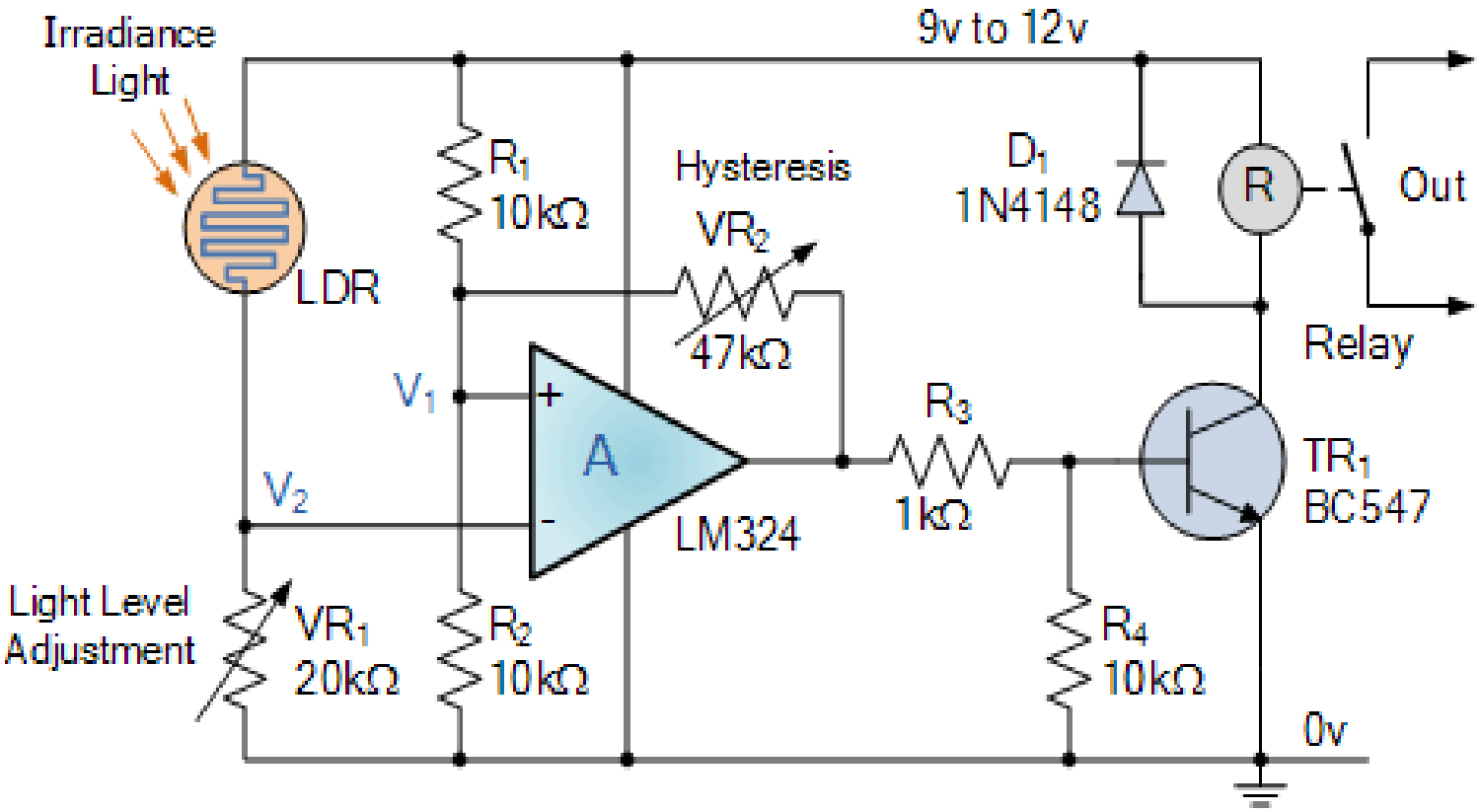
$$\therefore V_{out} = -V_1\left(\frac{R_3}{R_1}\right) + V_2\left(\frac{R_4}{R_2 + R_4}\right)\left(\frac{R_1 + R_3}{R_1}\right)$$

When resistors, R1 = R2 and R3 = R4 the above transfer function for the differential amplifier can be simplified to the following expression:

$$V_{OUT} = \frac{R_3}{R_1}\left(V_2 - V_1\right)$$

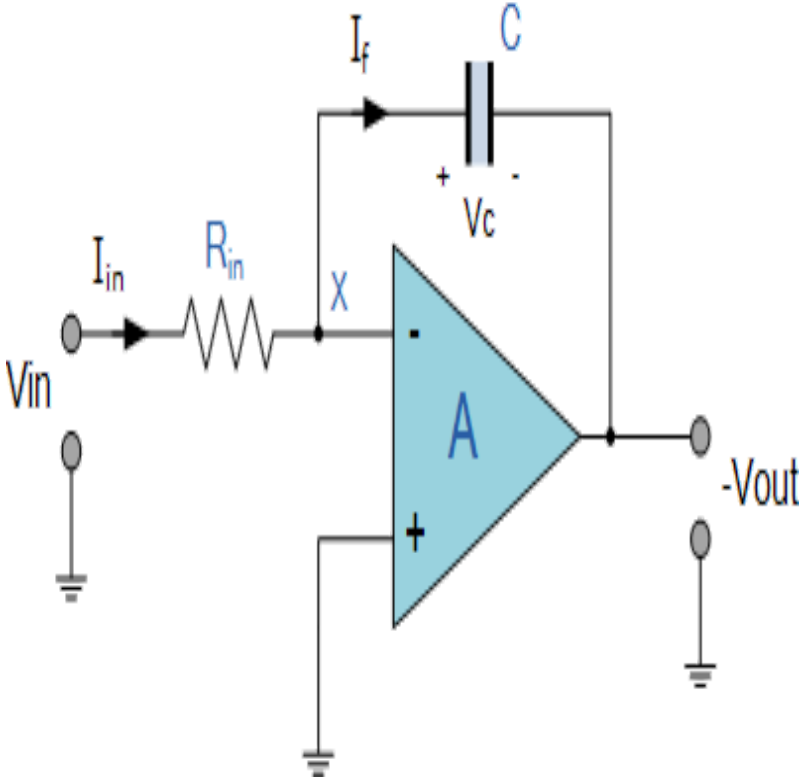


The standard Differential Amplifier circuit now becomes a differential voltage comparator by “Comparing” one input voltage to the other. For example, by connecting one input to a fixed voltage reference set up on one leg of the resistive bridge network and the other to either a “Thermistor” or a “Light Dependant Resistor” the amplifier circuit can be used to detect either low or high levels of temperature or light as the output voltage becomes



Circuit above acts as a light-activated switch which turns the output relay either “ON” or “OFF” as the light level detected by the LDR resistor exceeds or falls below some pre-set value. A fixed voltage reference is applied to the non-inverting input terminal of the op-amp via the R1 – R2 voltage divider network.

Integrator



Output voltage which is proportional to the integral of the input voltage.

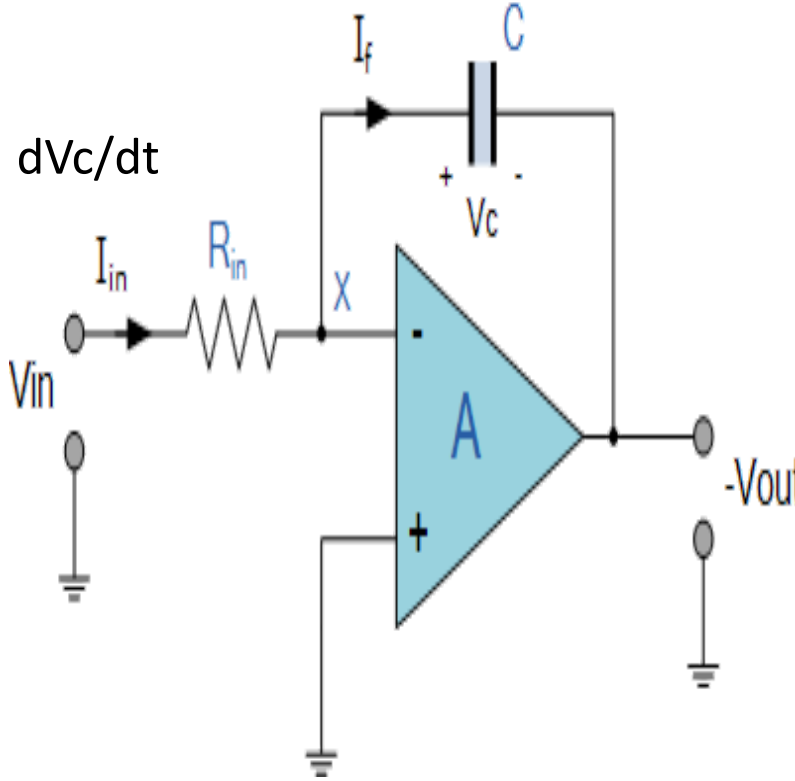
In other words the magnitude of the output signal is determined by the length of time a voltage is present at its input as the current through the feedback loop charges or discharges the capacitor as the required negative feedback occurs through the capacitor. The capacitor charges up at a rate determined by the RC

Integrator

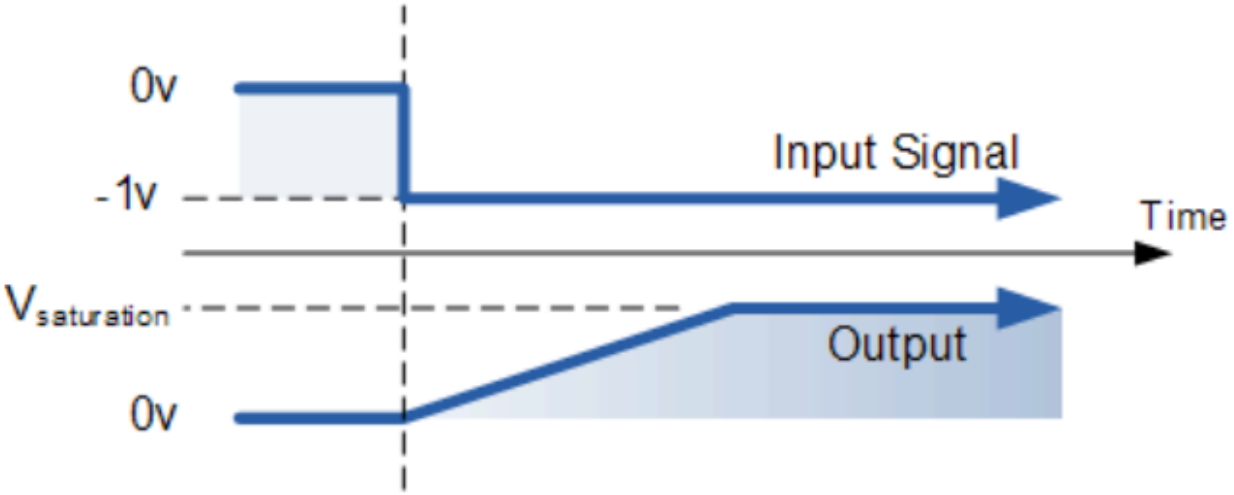
- $Ic = C \frac{d}{dt} Vc$
- $Iin = Ic$
- $Vin/R = C \frac{d}{dt} Vc$
- $= C \frac{d}{dt} (0 - Vout)$

Q = C Vc

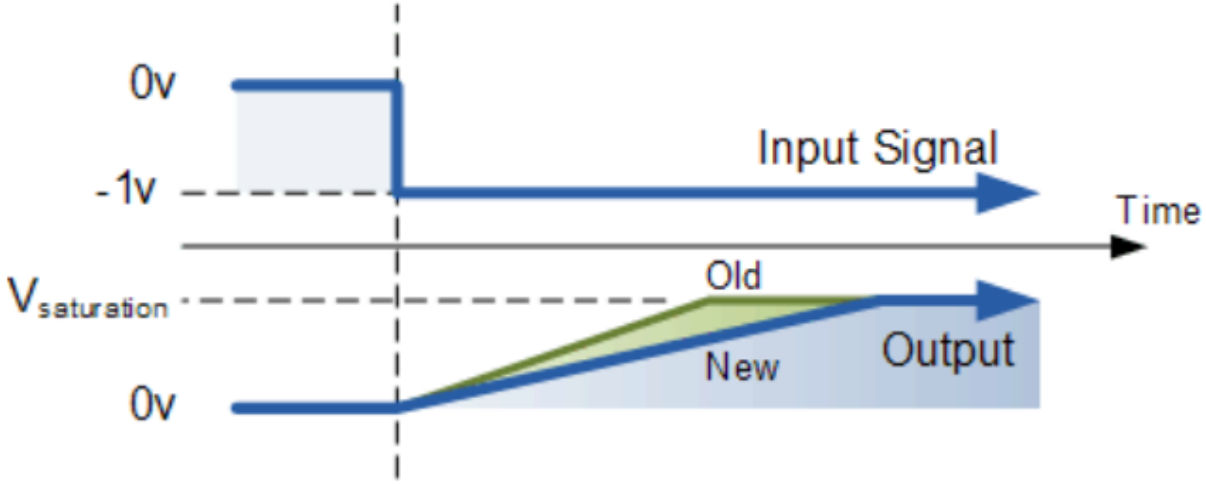
Ic = d Q/ dt = C dVc/dt



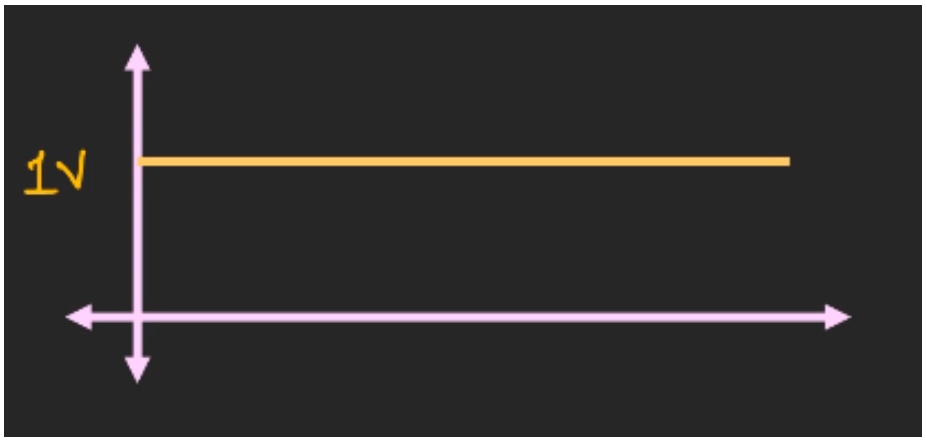
- $Vin/R = - C \frac{d}{dt} (Vout)$
- $\frac{d}{dt} (Vout) = (\frac{-1}{RC})\ Vin$
- $Vout = (\frac{-1}{RC}) \int Vin(t) dt$ + Vout (0^+)



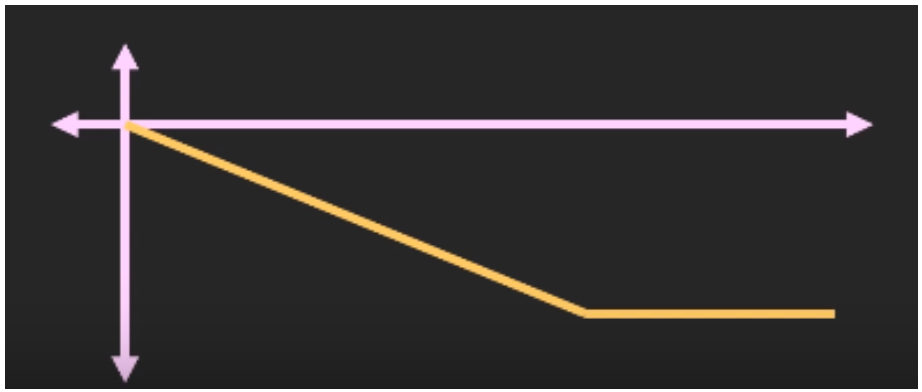
The rate at which the output voltage increases (the rate of change) is determined by the value of the resistor and the capacitor, "RC time constant". By changing this RC time constant value, either by changing the value of the Capacitor, C or the Resistor, R, the time in which it takes the output voltage to reach saturation can also be changed



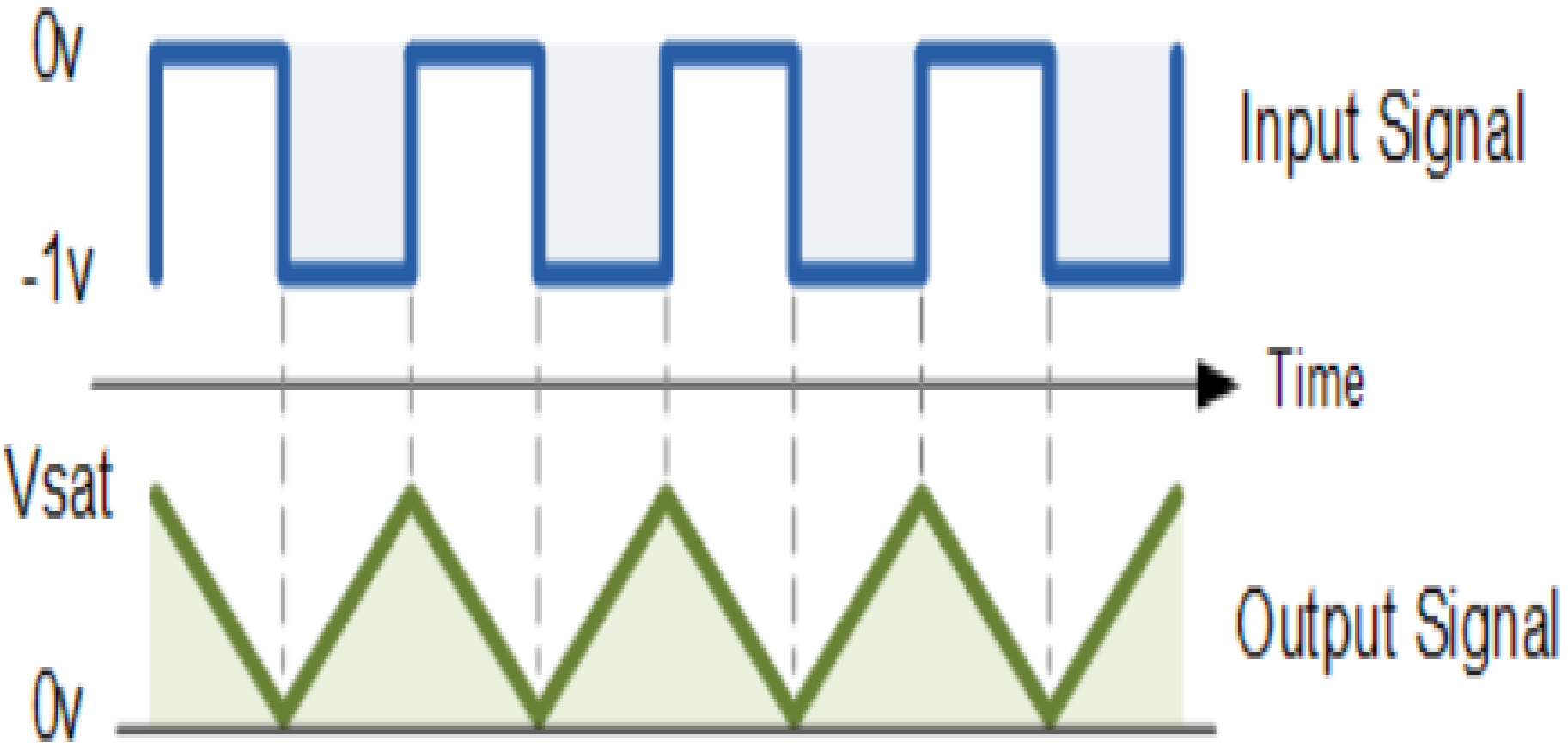
Integrator input, output



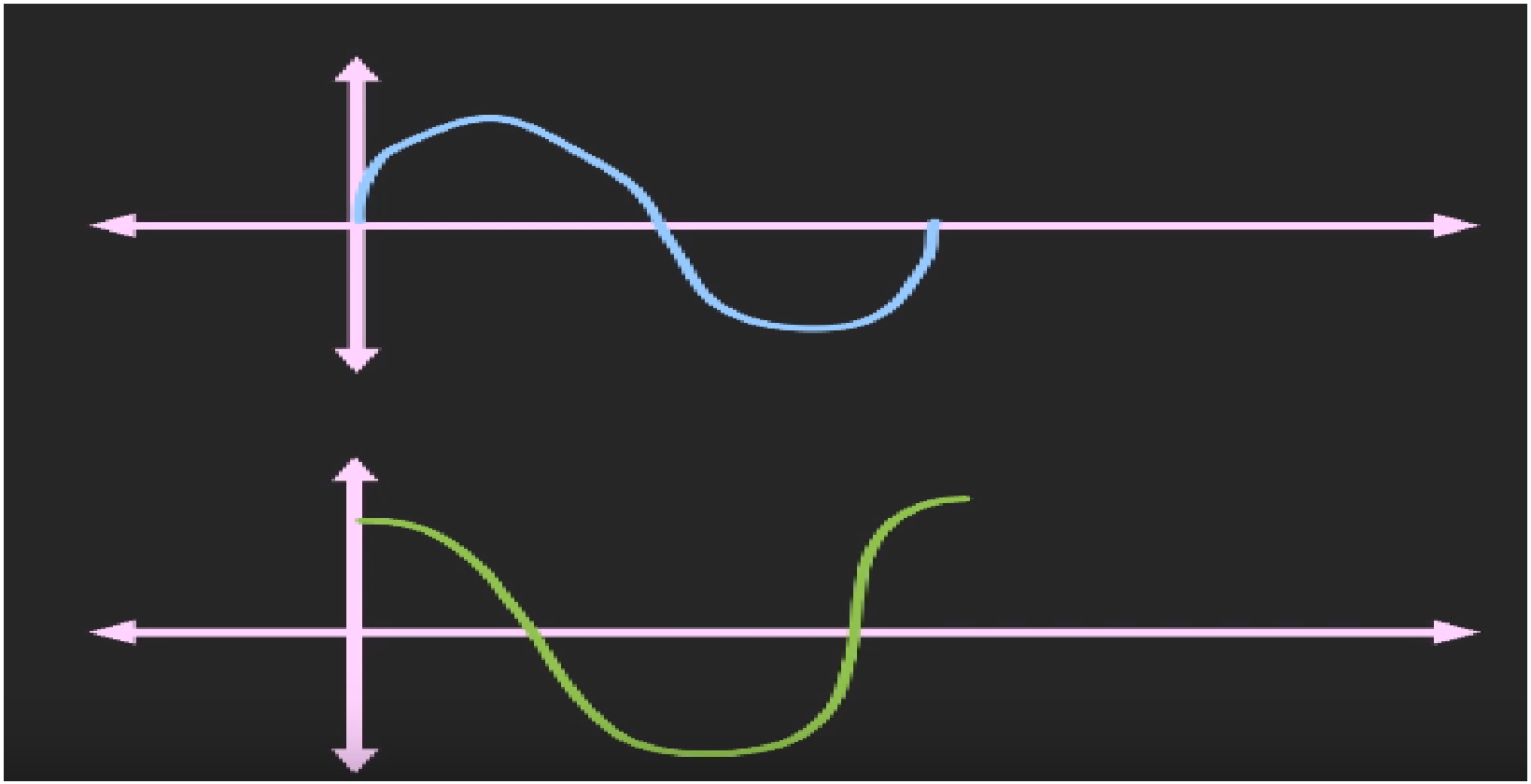
Vin



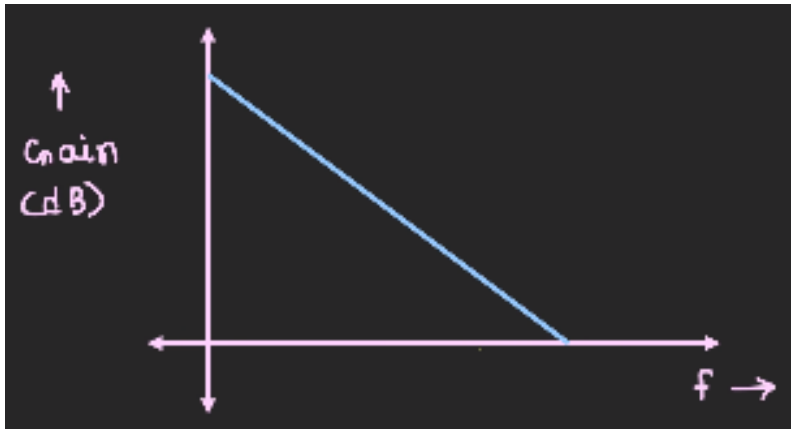
Vout


0v
-1v
Input Signal
Time
Vsat
0v
Output Signal

Example

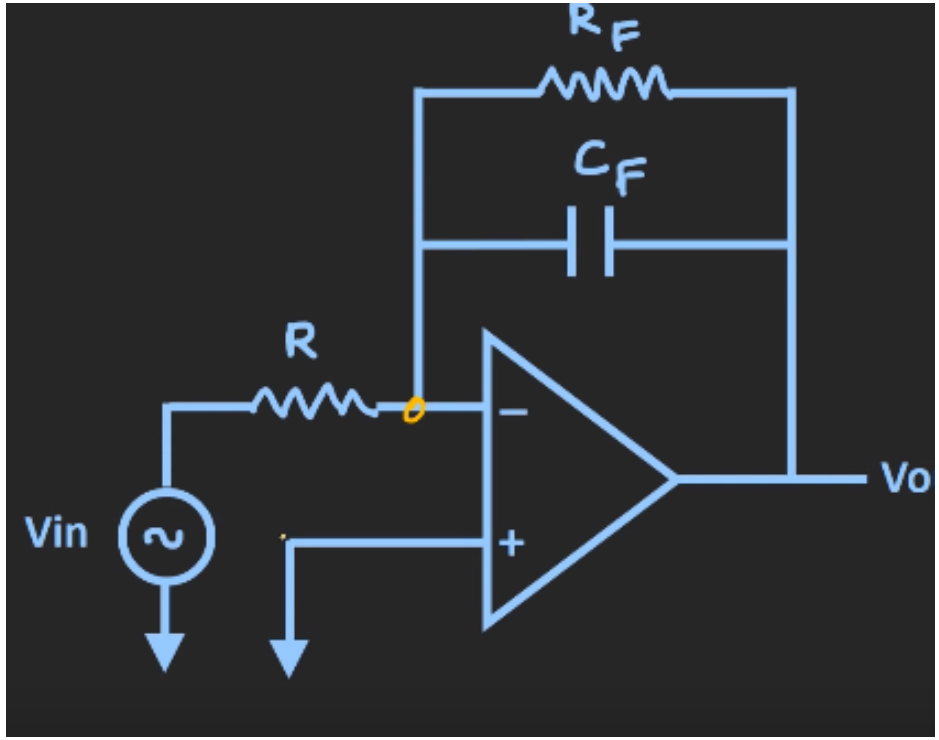


- Vout = - $\frac{Xc}{R}$ Vin
- A = - $\frac{Vout}{Vin}$ = - $\frac{Xc}{R}$ = - $\frac{1}{2\pi fRC}$
- Frequency response of Ideal Integrator

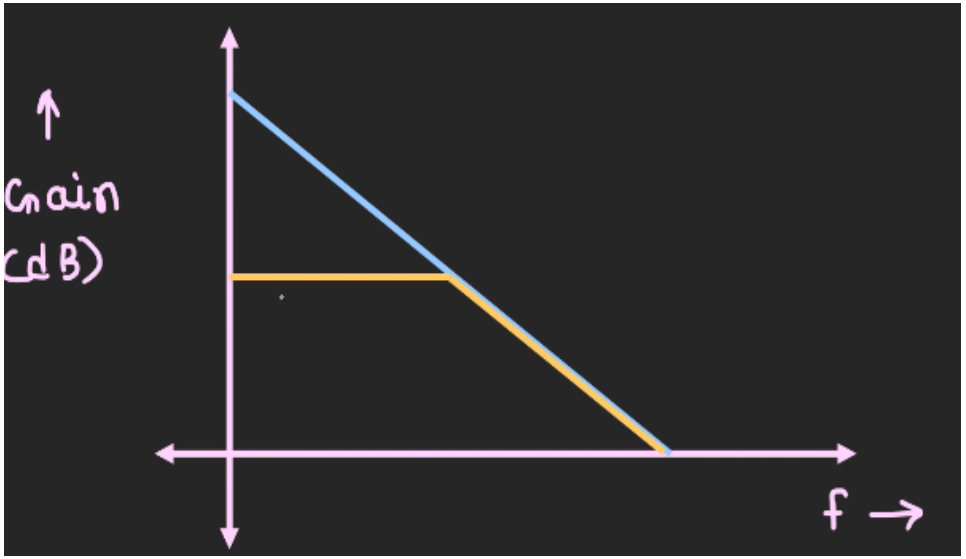


- Practically think about input offset voltage
- Even at mid frequency I/p offset voltage will get amplified and op-amp will move into saturation

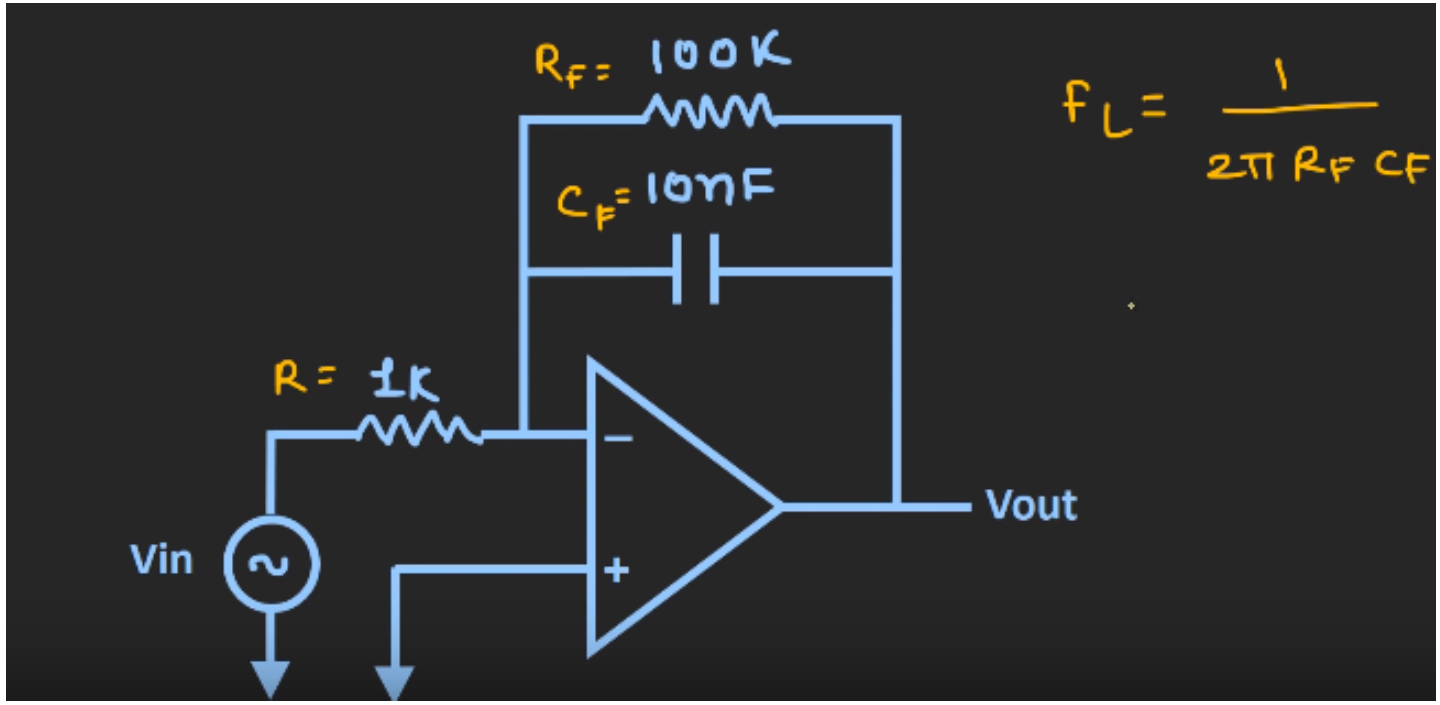
Practical Integrator



At low frequencies of the input signal, the circuit behaves normally like an integrator. At high frequencies, the capacitor acts as a short circuit and by-passes the resistor R_2.
The capacitor's reactance in turn reduces the gain of the amplifier.



Find lower cutoff frequency of Integrator



FL = 159 Hz

Fs > 10 FL

Integrator

- When a step voltage, Vin is first applied to the input of an integrating amplifier, the uncharged capacitor C has very little resistance and acts a bit like a short circuit allowing maximum current to flow via the input resistor, Rin as potential difference exists between the two plates. No current flows into the amplifiers input and point X is a virtual earth resulting in zero output. As the impedance of the capacitor at this point is very low, the gain ratio of Xc/Rin is also very small giving an overall voltage gain of less than one,(voltage follower circuit).
- As the feedback capacitor, C begins to charge up due to the influence of the input voltage, its impedance Xc slowly increase in proportion to its rate of charge. The capacitor charges up at a rate determined by the RC time constant, (τ) of the series RC network. Negative feedback forces the op-amp to produce an output voltage that maintains a virtual earth at the op-amp's inverting input

Integrator

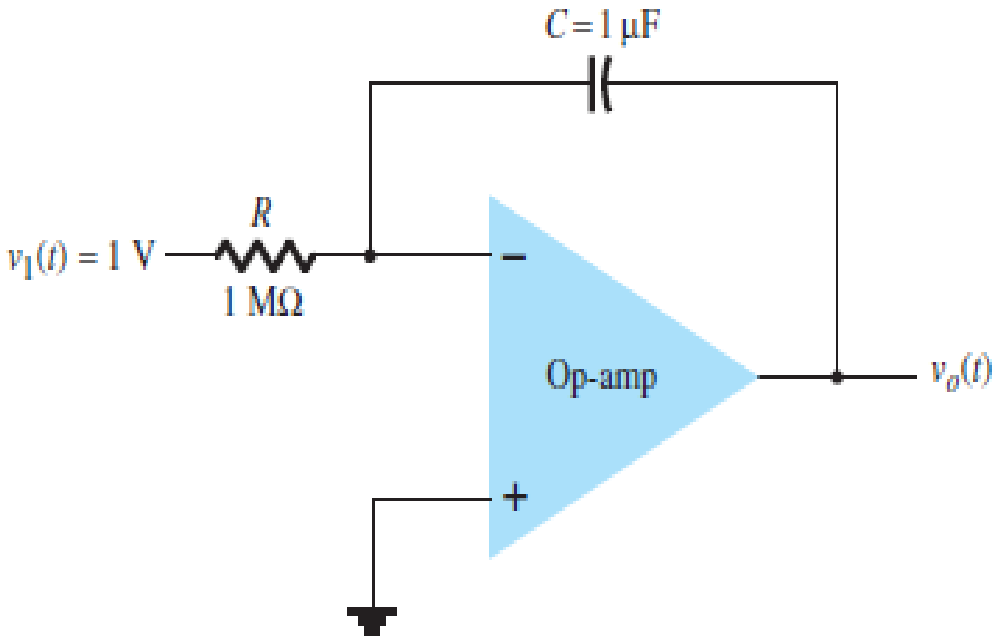
- Since the capacitor is connected between the op-amp's inverting input (which is at earth potential) and the op-amp's output (which is negative), the potential voltage, Vc developed across the capacitor slowly increases causing the charging current to decrease as the impedance of the capacitor increases. This results in the ratio of Xc/ Rin increasing producing a linearly increasing ramp output voltage that continues to increase until the capacitor is fully charged.
- At this point the capacitor acts as an open circuit, blocking any more flow of DC current. The ratio of feedback capacitor to input resistor (Xc/Rin) is now infinite resulting in infinite gain. The result of this high gain (similar to the op-amps open-loop gain), is that the output of the amplifier goes into saturation as shown below. (Saturation occurs when the output voltage of the amplifier swings heavily to one voltage supply rail or the other with

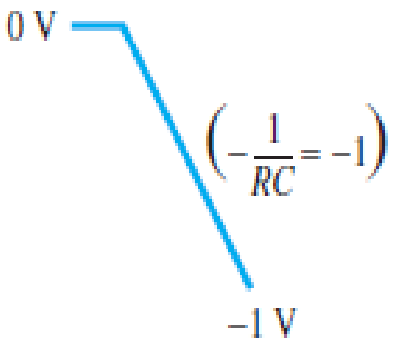
Integrator

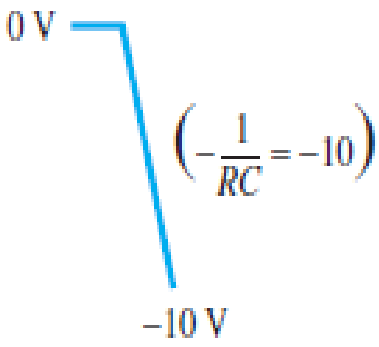
$$V_c = \frac{Q}{C}, \qquad V_c = V_x - V_{out} = 0 - V_{out}$$

$$\therefore \; -\frac{dV_{out}}{dt} = \frac{dQ}{Cdt} = \frac{1}{C}\frac{dQ}{dt}$$

Integrator

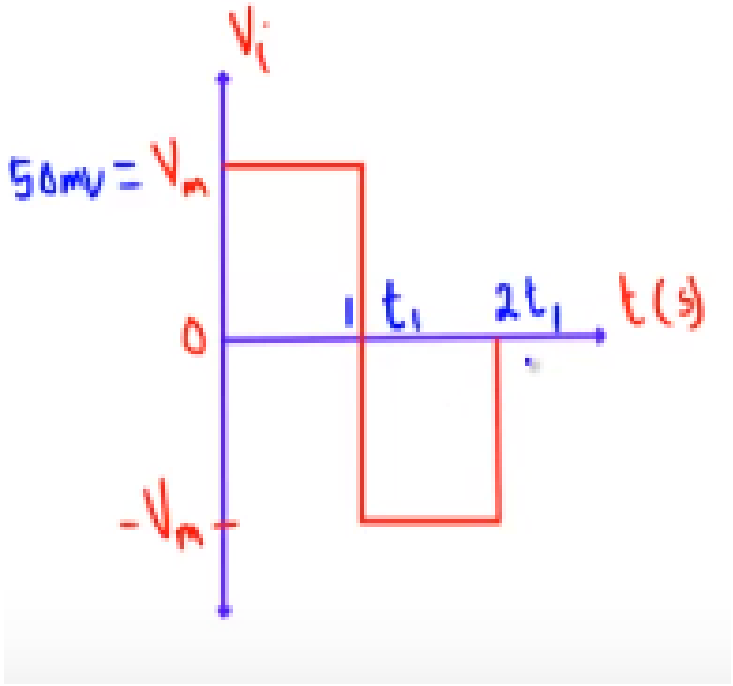


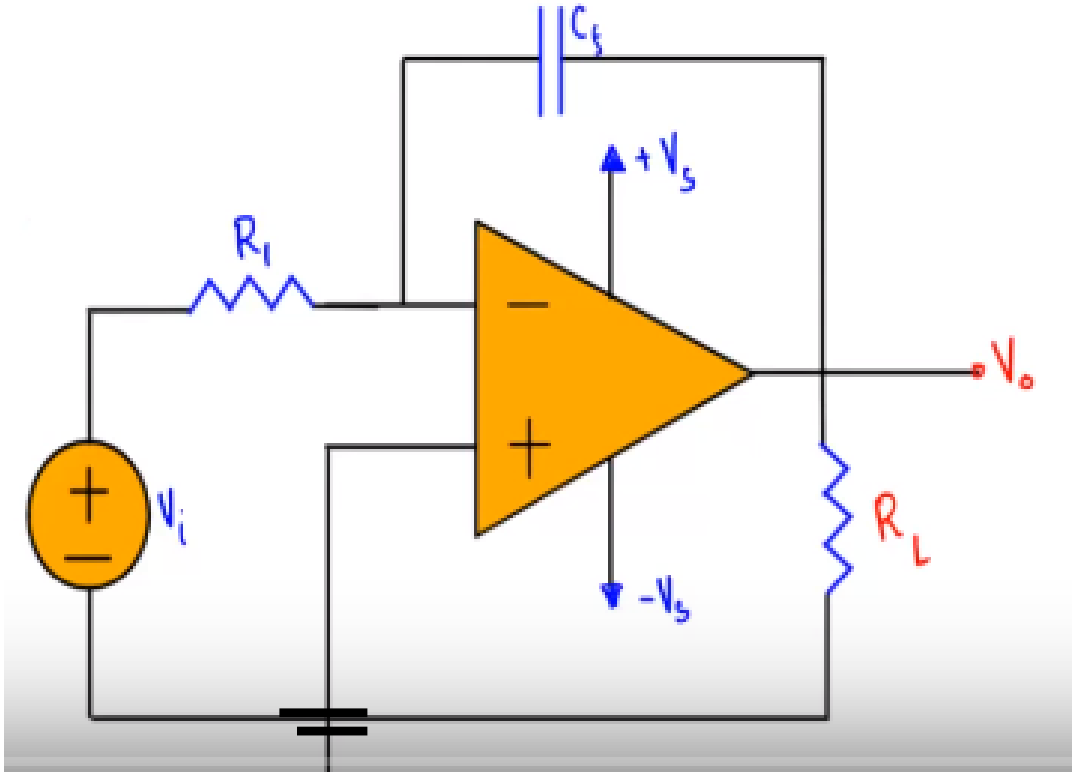


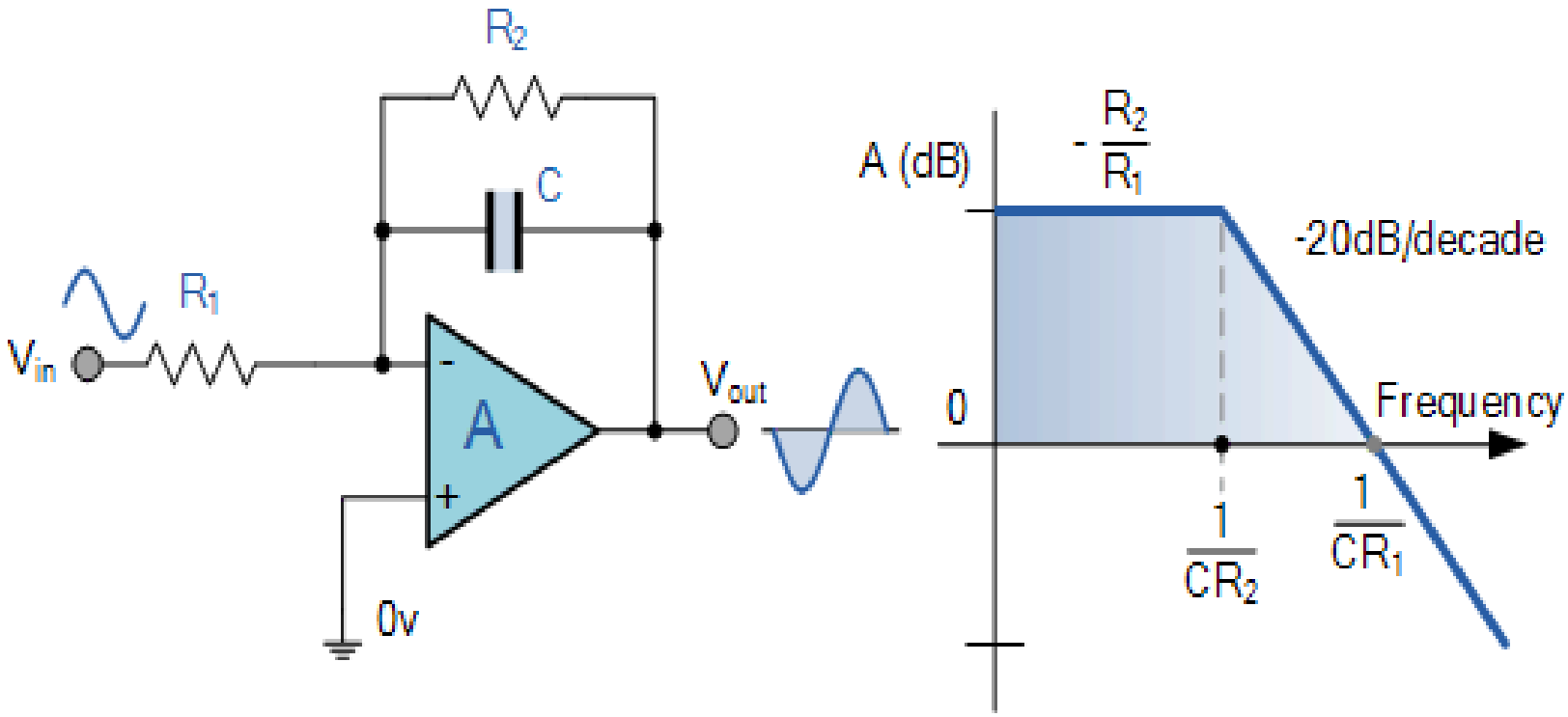


$$-\frac{1}{RC} = \frac{1}{(1\ \text{M}\Omega)(1\ \mu\text{F})} = -1$$

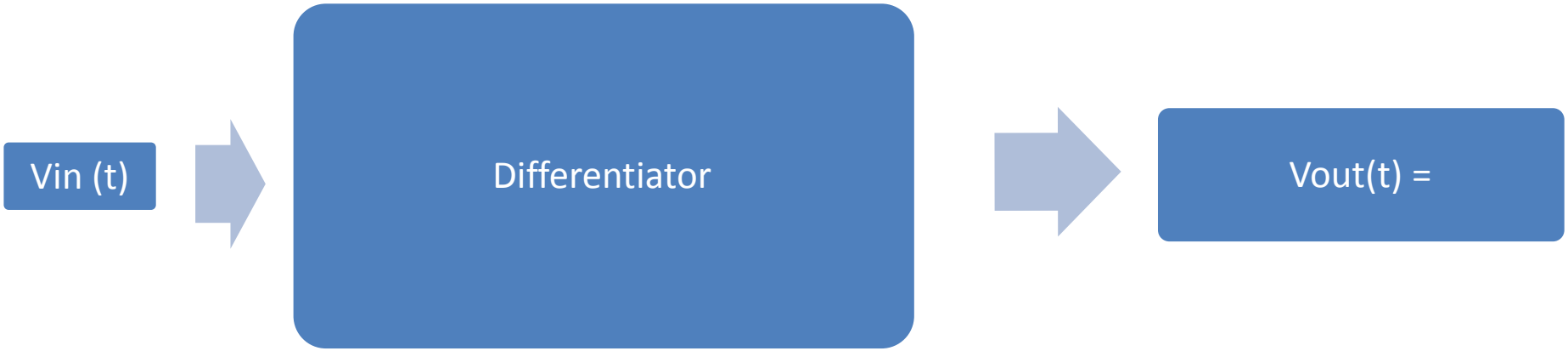
Vm = 50 mV; t1 = 1 Sec; R1 =100KΩ; Cf = 0.1 µF





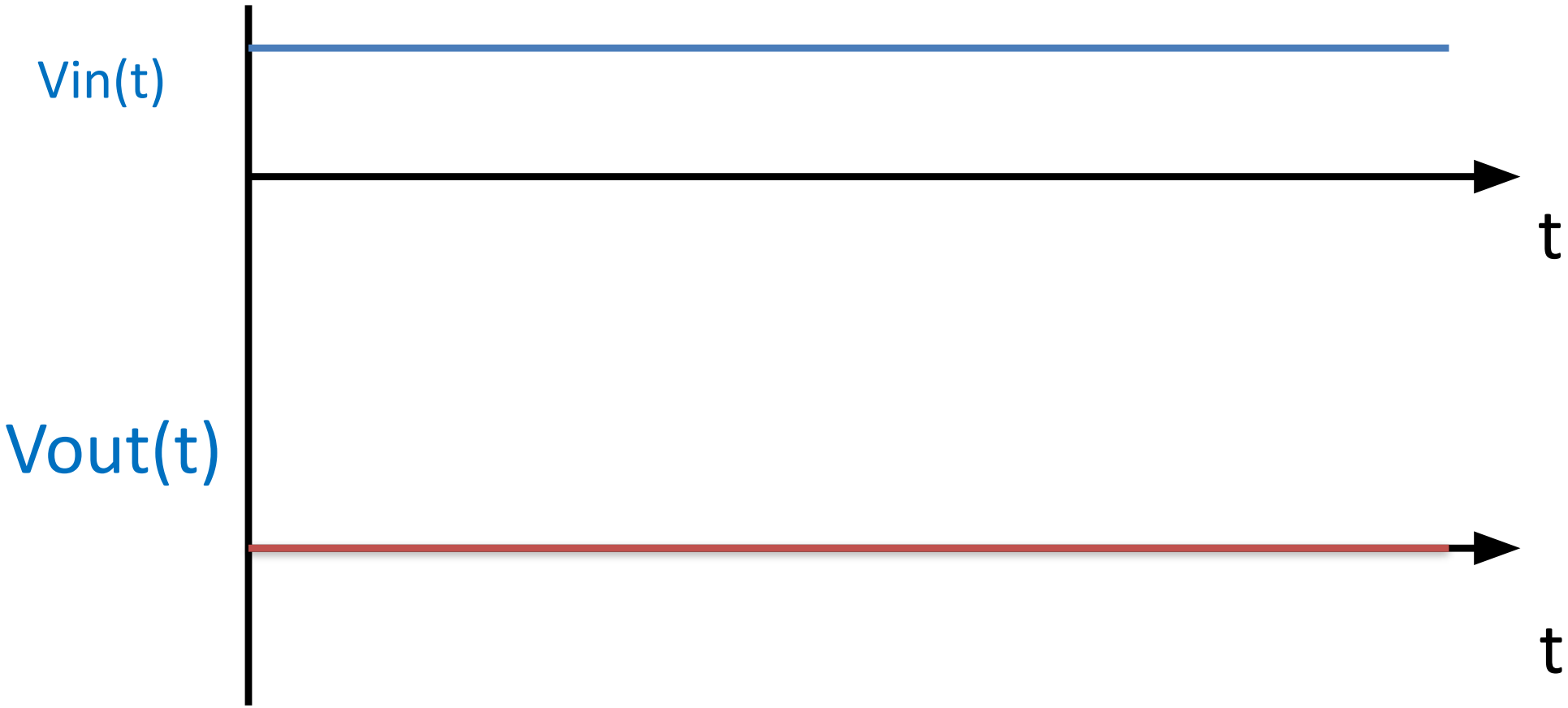

R2
C
R1
Vin
A
Vout
0v
A (dB)
-R2/R1
-20dB/decade
0
Frequency
1/CR2
1/CR1

Differentiator

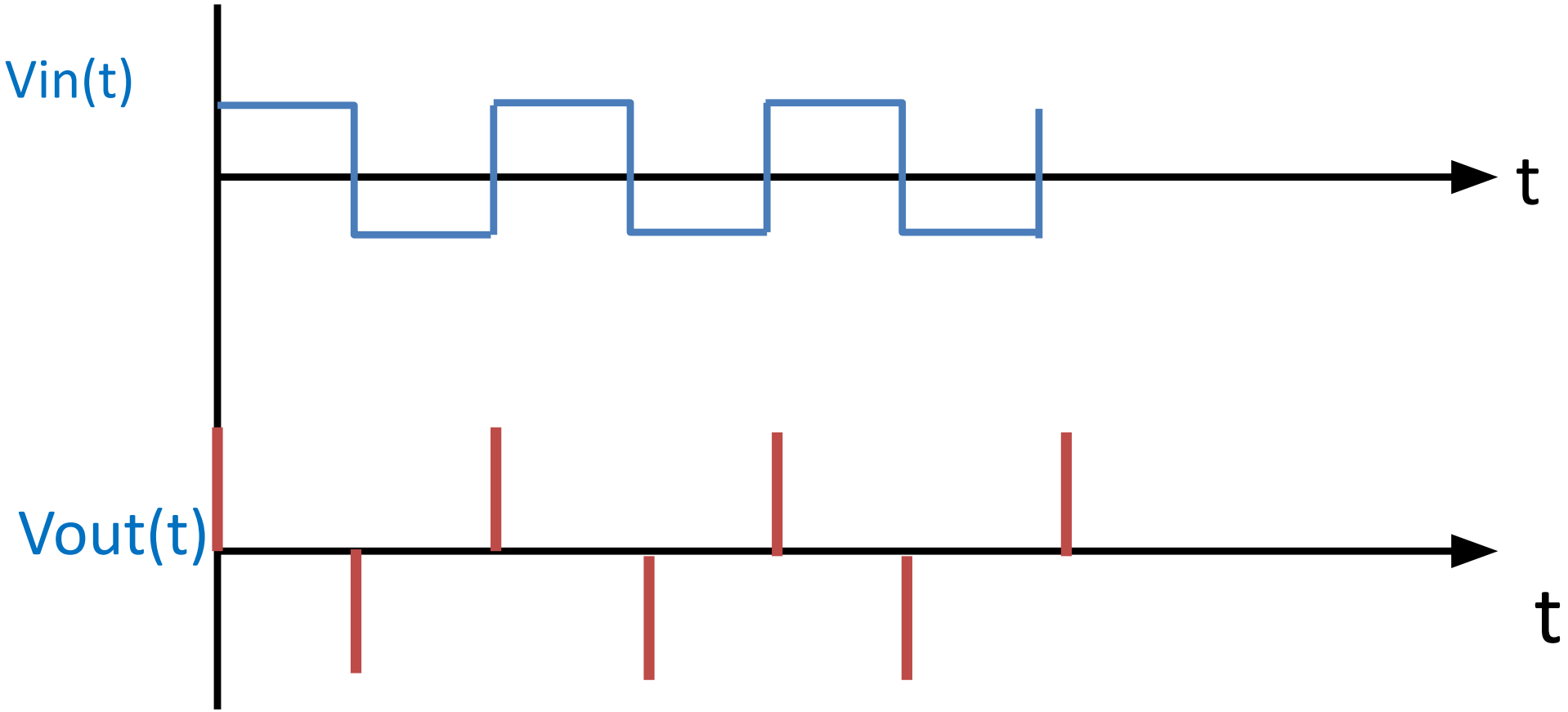


Using differentiator , we can get idea about rate at which input is changing

Differentiator



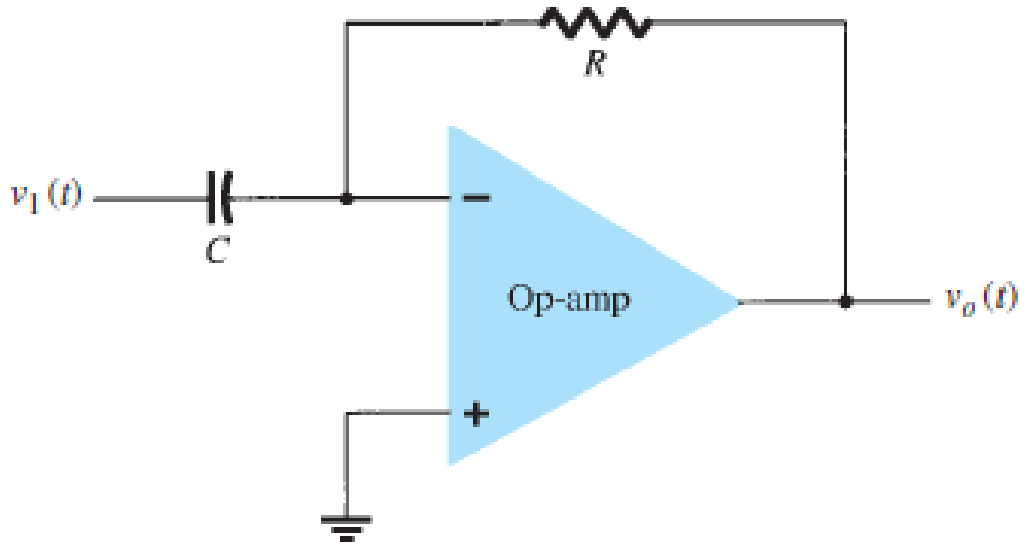
Differentiator



For Edge Detection..... to find High frequency component

Differentiator

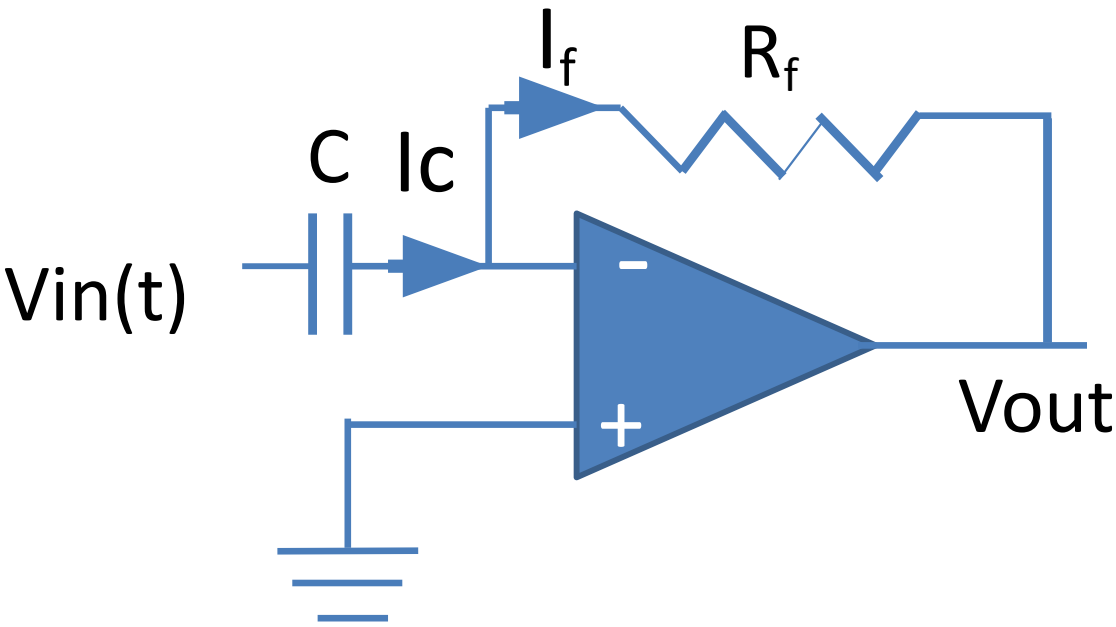
$$v_o(t) = -RC\frac{dv_1(t)}{dt}$$



Differentiator

The basic single resistor and single capacitor op-amp differentiator circuit is not widely used to reform the mathematical function of Differentiation because of the two inherent faults mentioned above, “Instability” and “Noise”. So in order to reduce the overall closed-loop gain of the circuit at high frequencies, an extra resistor, Rin is added

Differentiator



$I_c = I_f$

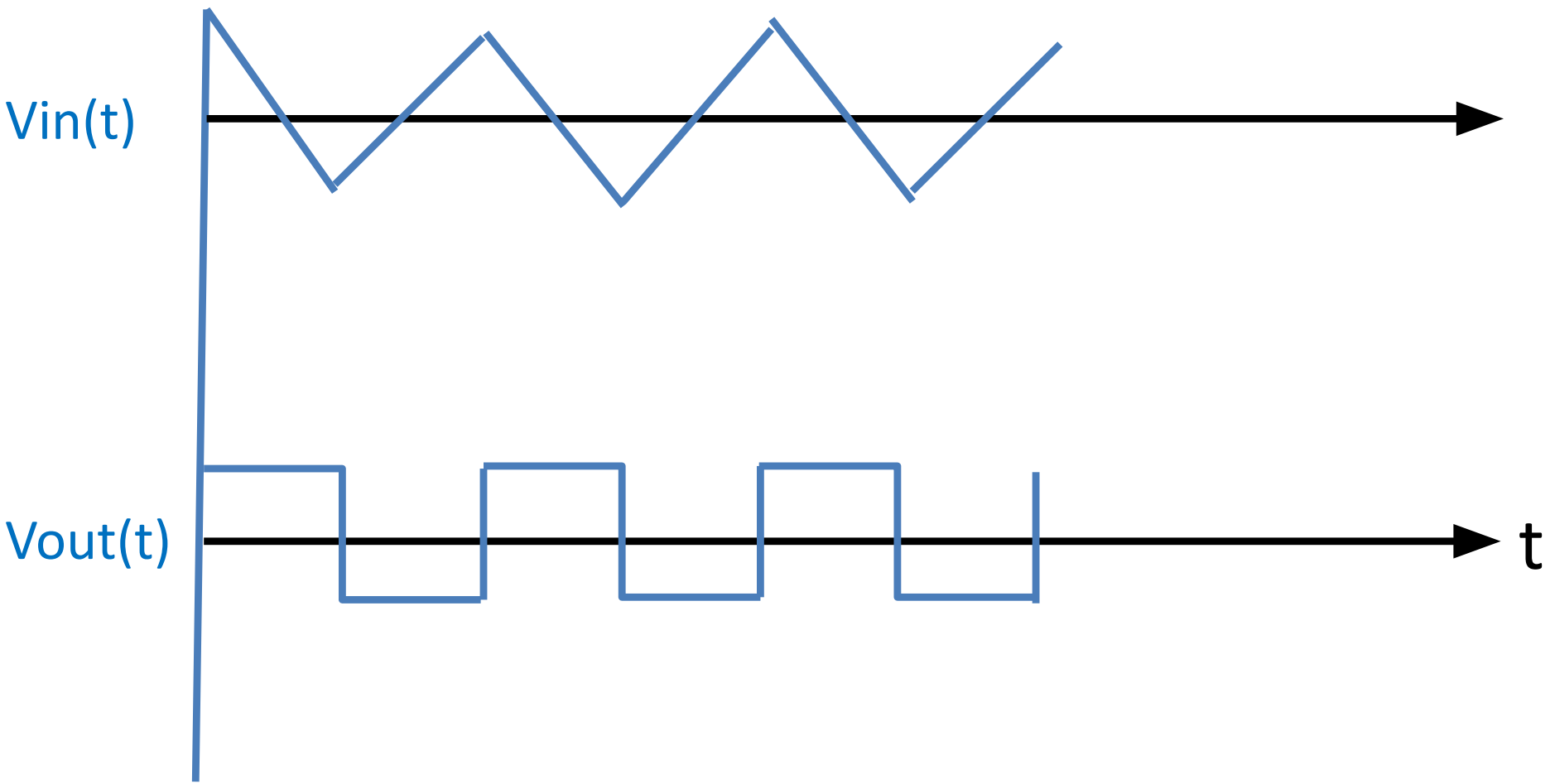
$Ic = C \frac{d}{dt} Vc$

$C \frac{d}{dt} Vc = (0\text{-}Vout)/ R_f$

$C \frac{d}{dt} Vin = -Vout/R_f$

$Vout = - R_f C \frac{d}{dt} Vin$

Differentiator



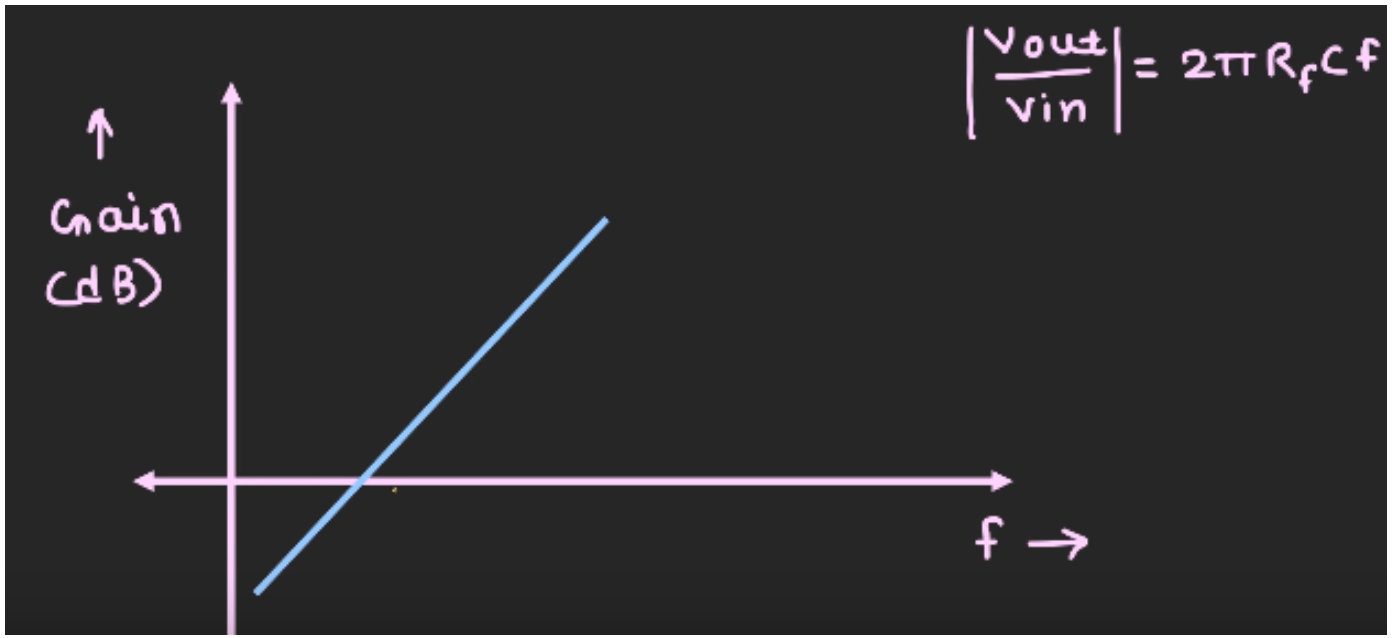
Limitations of differentiator

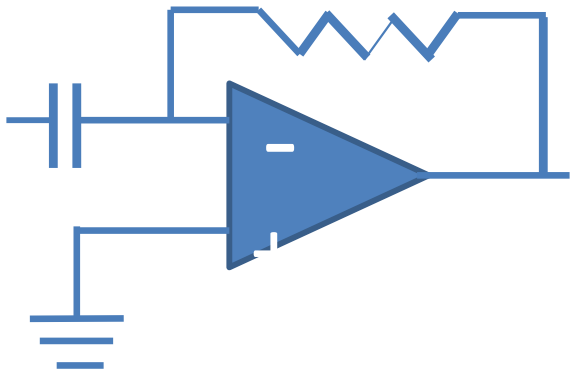
- **Noise Amplification**: The gain of an ideal differentiator is proportional to frequency ($Av = -j\omega RfCin$). This means that high-frequency noise, which is almost always present in electronic systems, is amplified much more than the desired signal. This can overwhelm the output and make the circuit unusable.
- **High-Frequency Instability**: At very high frequencies, the phase shift introduced by the differentiator circuit can combine with the op-amp's internal phase shifts. This can cause the total phase shift to reach 180°, turning the negative feedback into positive feedback and causing the circuit to oscillate.

Limitations of differentiator

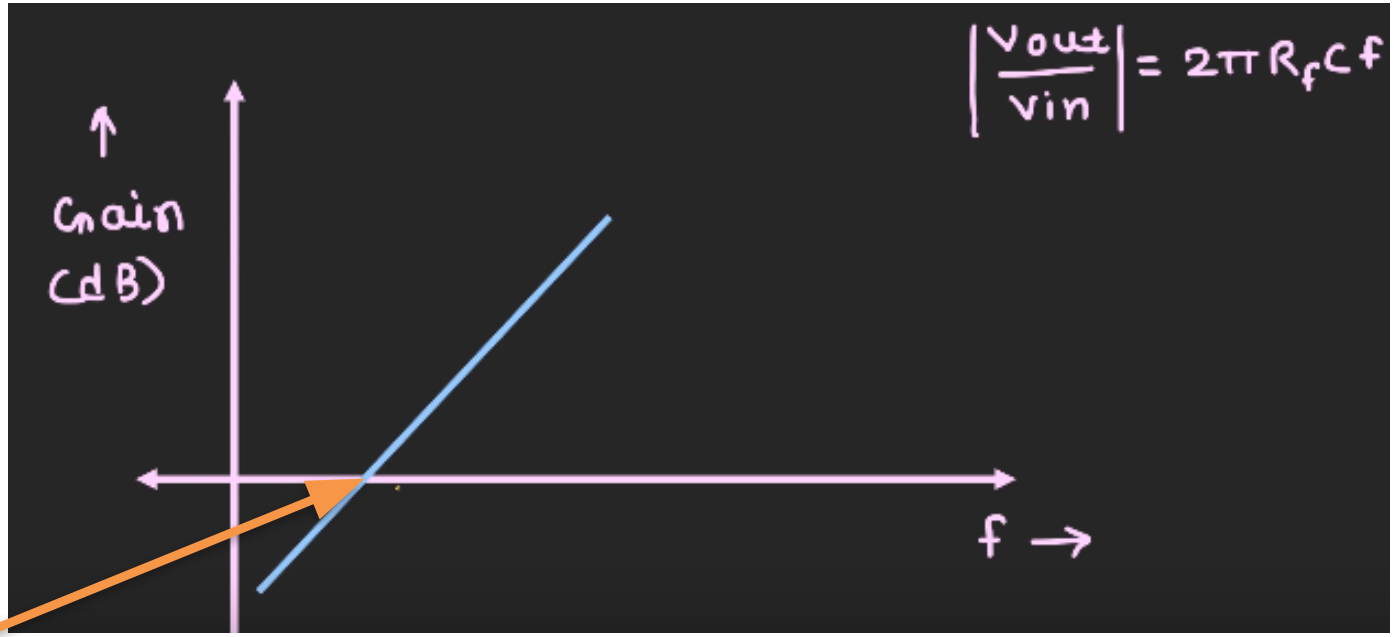
- At low frequency , input impedance will be very low.. Zin = Xc
- As frequency increases, gain will also increase. Differentiator is very sensitive to high frequency noise.

How to avoid these problems?





What will be value of 0 dB frequency?

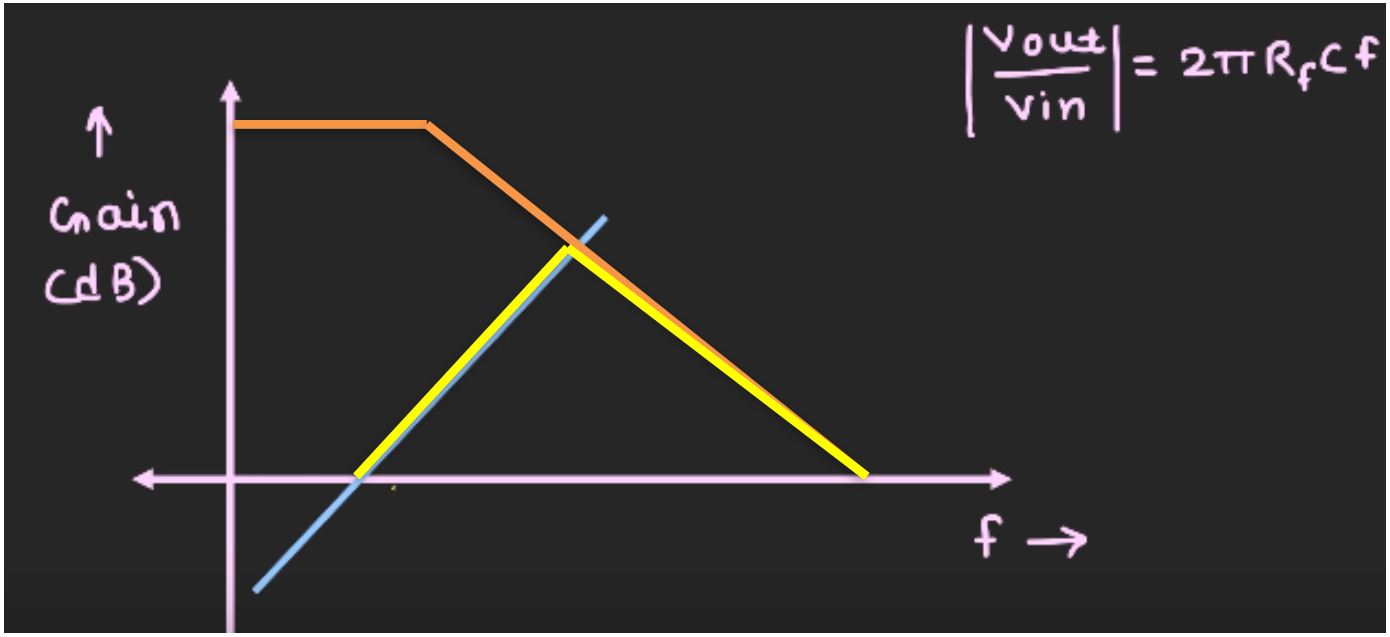


$$F_0 = \frac{1}{2\pi R_f C}$$

Can gain increase indefinitely with frequency?

No

It will be limited by open loop gain of op-amp



Gain of differentiator will be limited by open loop gain response of op-amp

Maximum gain attained by differentiator is intersection point of differentiator response and op-amp response

Limitations of differentiator

- At low frequency , input impedance will be very low.. Zin = Xc
- As frequency increases, gain will also increase. Differentiator is very sensitive to high frequency noise.

How to avoid these problems?

Add resistor in series with capacitor.

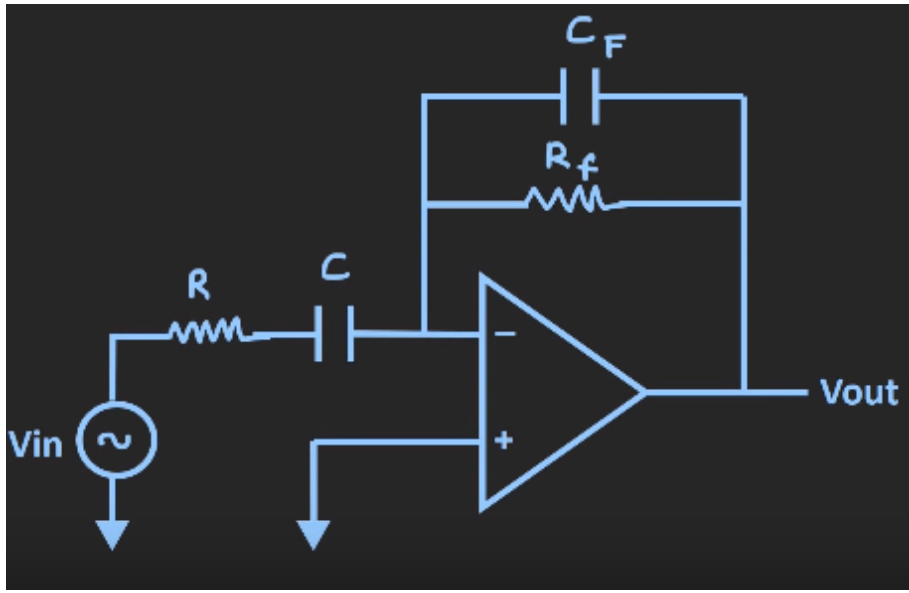
This will avoid problem of very low impedance at low frequency and high frequency sensitivity.

Zin = $\sqrt{R^2 + {X_c}^2}$

Zin $\geq \quad R$

High frequency gain = $\dfrac{-R_f}{R}$

By inserting C_f in parallel to R_f stability can be improved.



Why resistor in feedback?

- **For Integrators**
- A resistor is added in parallel with the feedback capacitor in a practical op-amp integrator circuit to prevent the op-amp from saturating due to its very high DC gain.
- At DC (zero frequency), a capacitor acts as an open circuit. Without the parallel resistor, the op-amp's gain at DC would be its **open-loop gain**, which is extremely high (often > 100,000). Any small DC input offset voltage or bias current would be massively amplified, causing the output to quickly "saturate" or "rail" to the positive or negative power supply voltage. The parallel resistor, typically a large value, provides a DC feedback path that limits the DC gain to a finite value, preventing this saturation.
- However, this resistor also makes the circuit a "leaky" or **lossy integrator**. At higher frequencies, the impedance of the feedback capacitor becomes much lower than the resistor's value, and the circuit behaves as an ideal integrator with the input capacitor. This limits the high-frequency gain, preventing the excessive amplification of noise and ensuring the circuit's stability.

Why resistor in feedback?

- **For Differentiators**
- A resistor is added in series with the input capacitor in a practical op-amp differentiator circuit to prevent high-frequency instability and noise amplification.
- An ideal op-amp differentiator has a gain that increases linearly with frequency
- This characteristic makes the circuit a high-pass filter.
- **Noise Amplification**: High-frequency electrical noise, which is almost always present in a circuit, will be heavily amplified, potentially overwhelming the desired signal and making the output unusable.
- **Instability**: At very high frequencies, the phase shift introduced by the op-amp's internal circuitry combined with the differentiator's phase shift can lead to a total phase shift of 180°, causing the negative feedback to become positive feedback. This can lead to unwanted oscillations, making the circuit unstable.
- The addition of the series resistor creates a low-pass filter in conjunction

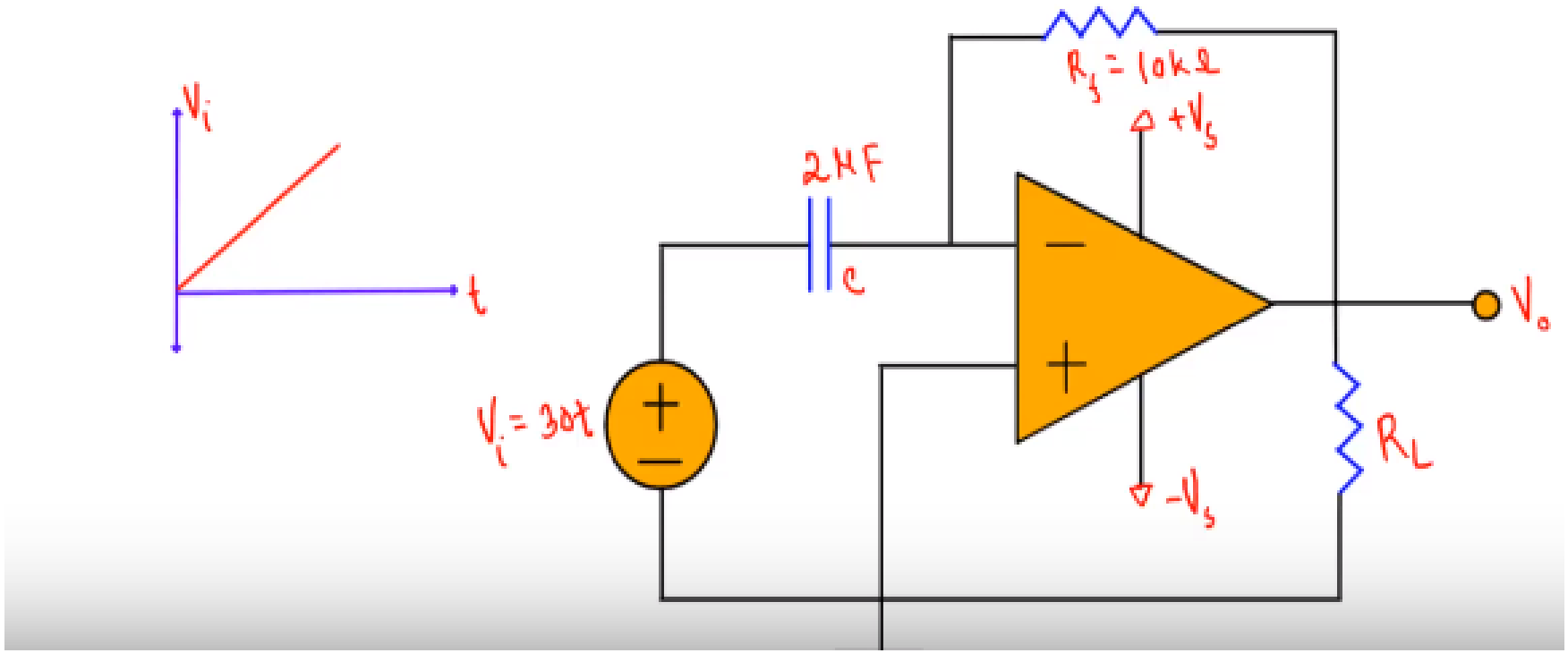
Example :Differentiator

Reference:

Fundamental of Electric Circuits - Alexander & Sadiku

Practice Problem - 6.14 (Third Edition)

The differentiator circuit has R = 10 kΩ, C = 2 μF. Given that, $V_i = 30t$, determine and draw the output.

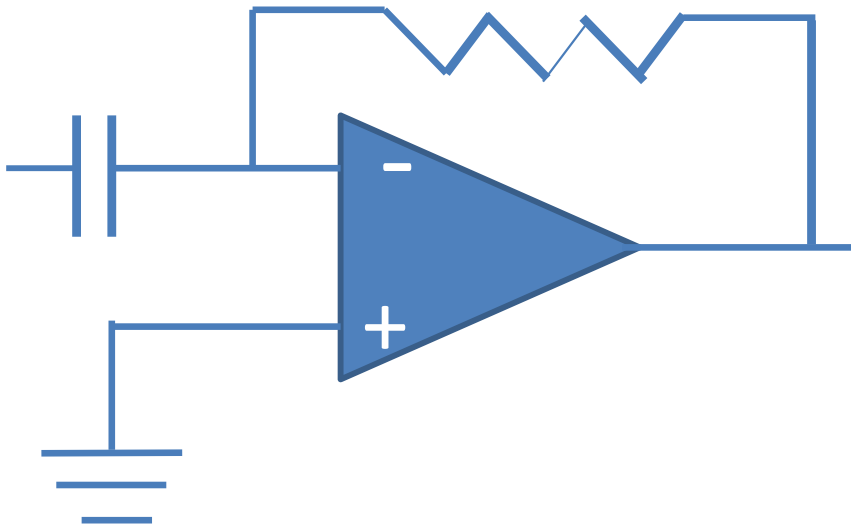


- $V_{out} = -R_f C \frac{d}{dt} V_{in}$

  $= - 20 \times 10^{-3} \times 30$

  $= - 0.6\ V$

Limitations of simple Differentiator

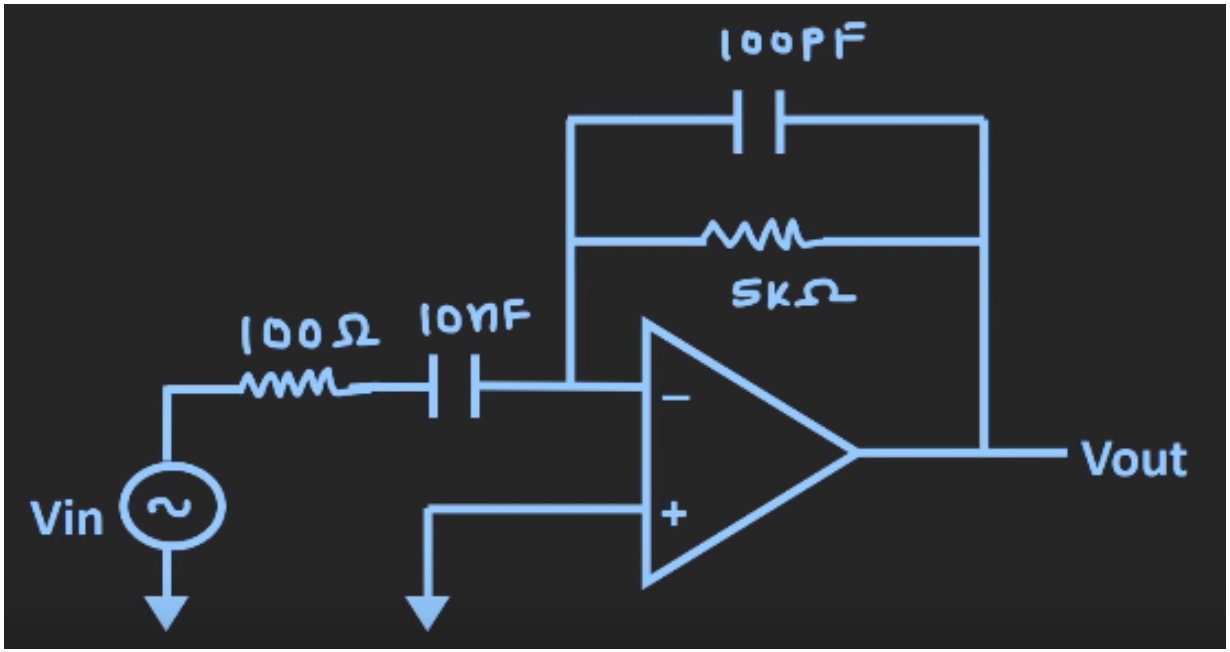


$$Vout = \frac{-R_f}{Xc} Vin$$

$$\frac{Vout}{Vin} = A = -R_f 2\pi f C$$

F ↑ A ↑

- Determine useful range of differentiation for the circuit



$$F_1 = \frac{1}{2\pi RfCin}$$

= 159.2 /3.18 kHz

$$F_2 = \frac{1}{2\pi RinC_F}$$

=318.3 KHz / 15.91MHz

- Offset Voltage, (V_{io})

Zero - The amplifiers output will be zero when the voltage difference between the inverting and the non-inverting inputs is zero, the same or when both inputs are grounded. Real op-amps have some amount of output offset voltage.

Open Loop Gain, (Avo)

Infinite - The main function of an operational amplifier is to amplify the input signal and the more open loop gain it has the better. Open-loop gain is the gain of the op-amp without positive or negative feedback and for an ideal amplifier the gain will be infinite but typical real values range from about 20,000 to 200,000.

- Input impedance, (Z_{in})

Infinite - Input impedance is the ratio of input voltage to input current and is assumed to be infinite to prevent any current flowing from the source supply into the amplifiers input circuitry (Iin =0). Real op-amps have input leakage currents from a few pico-amps to a few milli-amps

- Output impedance, (Z_{out})

Zero - The output impedance of the ideal operational amplifier is assumed to be zero acting as a perfect internal voltage source with no internal resistance so that it can supply as much current as necessary to the load. This internal resistance is effectively in series with the load thereby reducing the output voltage available to the load. Real op-amps have output-impedance in the 100-20Ω range

- Bandwidth, (BW)

Infinite - An ideal operational amplifier has an infinite frequency response and can amplify any frequency signal from DC to the highest AC frequencies so it is therefore assumed to have an infinite bandwidth. With real op-amps, the bandwidth is limited by the Gain-Bandwidth product (GB), which is equal to the frequency where the amplifiers gain becomes unity.

Reference

- Op-Amps and Linear Integrated Circuit by Gayakwad, Ramakant A.
- *Electronic Circuit Analysis and Design by Donald* Neaman
- https://www.youtube.com/watch?v=OPvs7A554Rw
- https://www.youtube.com/watch?v=aU24RWIgJVs
- https://www.electronics-tutorials.ws/opamp/opamp_5.html
- https://www.youtube.com/watch?v=2DFIr6t1hbc